

(See exercise 2 below for a precise formula for $n(\gamma)$.)

PROOF. We need to compute $\text{tr} = \text{tr}_{\mathbb{Z}\Gamma/\mathbb{Z}\Gamma'}: T(\mathbb{Z}\Gamma) \rightarrow T(\mathbb{Z}\Gamma')$. Fix $\gamma \in \Gamma$, and recall that $\text{tr}([\gamma])$ is the trace over $\mathbb{Z}\Gamma'$ of the right multiplication map $\mu_\gamma: \mathbb{Z}\Gamma \rightarrow \mathbb{Z}\Gamma$ (cf. proof of 2.4). Let S be a set of coset representatives for $\Gamma' \backslash \Gamma$. Then S is a basis for $\mathbb{Z}\Gamma$ as a left $\mathbb{Z}\Gamma'$ -module, and μ_γ can be described as follows in terms of this basis: Given $s \in S$, we have $\mu_\gamma(s) = s\gamma = \gamma't$, where $t \in S$ is the representative of the coset $\Gamma's\gamma$ and $\gamma' = s\gamma t^{-1} \in \Gamma'$. Thus we get a non-zero contribution to the trace of μ_γ if and only if $t = s$, i.e., if and only if $s\gamma s^{-1} \in \Gamma'$, and this contribution is then $\gamma' = s\gamma s^{-1}$. We therefore have

$$\text{tr}([\gamma]) = \sum_{s \in S} [s\gamma s^{-1}]_{\Gamma'},$$

where

$$[s\gamma s^{-1}]_{\Gamma'} = \begin{cases} \text{the } \Gamma'\text{-conjugacy class of } s\gamma s^{-1} & \text{if } s\gamma s^{-1} \in \Gamma' \\ 0 & \text{otherwise.} \end{cases}$$

For any $\gamma \in \Gamma'$, let $n(\gamma)$ be the number of cosets $\Gamma's$ such that $[s\gamma s^{-1}]_{\Gamma'} = [\gamma]_{\Gamma'}$. Note that $n(\gamma) \geq 1$ and that $n(1) = (\Gamma: \Gamma')$. Let $\mathcal{C}$ (resp. $\mathcal{C}'$) be a set of representatives for the Γ -conjugacy classes (resp. Γ' -conjugacy classes). Then the formula of the previous paragraph yields the following formula for $\text{tr}: T(\mathbb{Z}\Gamma) \rightarrow T(\mathbb{Z}\Gamma')$:

$$\text{tr}\left(\sum_{\gamma \in \mathcal{C}} r(\gamma)[\gamma]\right) = \sum_{\gamma \in \mathcal{C}'} n(\gamma)r(\gamma)[\gamma]_{\Gamma'}.$$

The proposition is now immediate from 2.4. □

For example, if we take Γ finite and $\Gamma' = \{1\}$, 4.1 simply says $\text{rk}_{\mathbb{Z}}(P) = |\Gamma| \cdot R_{\Gamma}(P)(1)$; thus the “rank” $\rho_{\Gamma}(P)$ introduced in §1 satisfies

$$(4.2) \quad \rho_{\Gamma}(P) = R_{\Gamma}(P)(1).$$

[Note: We have just rederived the result of exercise 6 of §2.] On the other hand, our other “rank” $\varepsilon_{\Gamma}(P)$ can also be expressed in terms of $R_{\Gamma}(P)$. Indeed, we have already done this in exercise 5 of §2, and the result of that exercise, when translated into the notation of the present section, is

$$(4.3) \quad \varepsilon_{\Gamma}(P) = \sum_{\gamma \in \mathcal{C}} R_{\Gamma}(P)(\gamma).$$

We can now prove our main result, which can be viewed as saying that the three “ranks” ε , ρ , and R that we have been talking about for finite Γ are all essentially the same:

(4.4) Theorem (Swan [1960b]). *If Γ is finite and P is a finitely generated projective $\mathbb{Z}\Gamma$ -module, then $R_{\Gamma}(P)(\gamma) = 0$ for $\gamma \neq 1$. Thus there is an integer r such that $R_{\Gamma}(P) = r \cdot [1]$, and one has $\varepsilon_{\Gamma}(P) = \rho_{\Gamma}(P) = r$.*

PROOF. The second assertion follows from the first in view of 4.2 and 4.3. To prove the first assertion, we may replace Γ by the cyclic subgroup generated by γ ; for the restriction formula 4.1 shows that this does not change the question as to whether $R_\Gamma(P)(\gamma) = 0$. In particular, we may assume that $\mathbb{Z}\Gamma$ is commutative. Also, as we pointed out in exercise 7 of §2, it follows from formal properties of $\rho_\Gamma(\)$ that $\mathbb{Z}\Gamma$ is indecomposable. Theorem 3.5 is therefore applicable and shows that $R_\Gamma(P)$ is an integral multiple of $[1]$, as required. $\square$

For emphasis, we go back to the definitions of ρ and ε and state explicitly what it means for them to be equal:

(4.5) Corollary. *Let Γ be a finite group and P a finitely generated projective $\mathbb{Z}\Gamma$ -module. Then*

$$\mathrm{rk}_{\mathbb{Z}}(P) = |\Gamma| \cdot \mathrm{rk}_{\mathbb{Z}}(P_\Gamma).$$

Remark. Using some elementary representation theory (cf. exercise 3 below), one can restate Swan's theorem as follows: If Γ is finite and P is a finitely generated projective $\mathbb{Z}\Gamma$ -module, then $\mathbb{Q} \otimes_{\mathbb{Z}} P$ is a free $\mathbb{Q}\Gamma$ -module. It is in this form that Swan stated and proved the theorem. The formulation and proof in terms of Hattori–Stallings ranks are due to Bass [1976, 1979]. Bass went on to conjecture that 4.4 remains valid for arbitrary groups Γ . [To make sense out of the second assertion of 4.4 for arbitrary Γ , one should take 4.2 as a *definition* when Γ is infinite, as we essentially did in exercise 8 of §2.] He proved the conjecture for a large class of *torsion-free* groups Γ . For infinite groups with torsion, however, the conjecture is still completely open.

EXERCISES

1. Where did the proof of 4.4 use the assumption that Γ was finite?
2. Show that the integer $n(\gamma)$ in 4.1 is equal to $(C_\Gamma(\gamma): C_\Gamma(\gamma))$, where $C_\Gamma(\gamma)$ (resp. $C_\Gamma(\gamma)$) is the centralizer of γ in Γ (resp. Γ).
3. Let Γ be finite, let k be a field of characteristic zero, and let V be a $k\Gamma$ -module which is finite dimensional over k . Then V is finitely generated and projective (cf. §I.8, exercise 5), hence it has a Hattori–Stallings rank $\sum R(\gamma) \cdot [\gamma]$, where $R: \Gamma \rightarrow k$ is a central function, i.e., a function which is constant on conjugacy classes. On the other hand, there is a classical way to use traces to associate a central function $\chi: \Gamma \rightarrow k$ to V , called the *character* of V . Namely, $\chi(\gamma)$ is the trace over k of the action of γ on V . The purpose of this exercise is to relate χ and R and to draw some consequences from this.

(a) Show that the Hattori–Stallings rank of V is equal to

$$\frac{1}{|\Gamma|} \sum_{\gamma \in \Gamma} \chi(\gamma^{-1}) \cdot [\gamma] = \sum_{\gamma \in \Gamma} \frac{\chi(\gamma^{-1})}{|C_\Gamma(\gamma)|} \cdot [\gamma].$$

i.e., that $R(\gamma) = \chi(\gamma^{-1})/|C_\Gamma(\gamma)|$. [Method 1: There is a canonical way of writing V as a direct summand of an induced module; namely, we have $k\Gamma \otimes_k V \xrightarrow{\sim} V$, where $\pi(\gamma \otimes v) = \gamma v$ and

$$\iota(v) = \frac{1}{|\Gamma|} \sum_{\gamma \in \Gamma} \gamma \otimes \gamma^{-1}v.$$

Choose a basis (e_i) for V over k , so that $(1 \otimes e_i)$ is a basis for $k\Gamma \otimes V$ over $k\Gamma$. You can now write down the matrix of $\iota\pi$ in terms of the matrices $a_{ij}(\gamma)$ of the elements of Γ acting on V . Method 2: It's enough to prove this when k is algebraically closed and V is irreducible. In this case, the element

$$e = \frac{n}{|\Gamma|} \sum_{\gamma \in \Gamma} \chi(\gamma^{-1})\gamma,$$

where $n = \dim_k V$, is the idempotent of $k\Gamma$ which projects $k\Gamma$ onto its summand of type V (cf. Serre [1977b], 2.6). Since V occurs n times in $k\Gamma$, the image of e in $T(k\Gamma)$ is the Hattori–Stallings rank of the sum of n copies of V . Method 3: By the restriction formula (in the precise form provided by exercise 2), we may assume Γ is cyclic, so that the map $\mu_\gamma: V \rightarrow V$ given by the action of γ is $k\Gamma$ -linear for each $\gamma \in \Gamma$. Then $\text{tr}_{k\Gamma}(\mu_\gamma) = \gamma \cdot \text{tr}_{k\Gamma}(\text{id}) = \gamma \cdot R_{k\Gamma}(V)$ by exercise 3 of §2. In particular, the coefficient $\text{tr}_{k\Gamma}(\mu_\gamma)(1)$ of $[1]$ is equal to $R_{k\Gamma}(V)(\gamma^{-1})$. On the other hand, the analogue of 4.1 for traces gives $\chi(\gamma) = \text{tr}_k(\mu_\gamma) = |\Gamma| \cdot \text{tr}_{k\Gamma}(\mu_\gamma)(1)$.]

(b) Using the fact (known from representation theory) that the module V is determined up to isomorphism by its character, show that two finitely generated $k\Gamma$ -modules are isomorphic if and only if they have the same Hattori–Stallings rank.

(c) Deduce the reformulation of 4.4 given in the remark above.

4. Take 4.2 as a definition, and prove that $\rho_{\Gamma'}(P) = (\Gamma: \Gamma')\rho_\Gamma(P)$ for any group Γ , subgroup Γ' of finite index, and finitely generated projective $\mathbb{Z}\Gamma$ -module P .
5. With Γ , Γ' , and P as in exercise 4, prove that $\varepsilon_{\Gamma'}(P) = (\Gamma: \Gamma')\varepsilon_\Gamma(P)$. [Hint: Suppose first that Γ' is normal in Γ . Let $\bar{\Gamma} = \Gamma/\Gamma'$ and let $\bar{P} = P_{\Gamma'} = \mathbb{Z}\bar{\Gamma} \otimes_{\mathbb{Z}\Gamma} P$ (cf. §II.2, exercise 3). $\bar{P}$ is a finitely generated projective $\mathbb{Z}\bar{\Gamma}$ -module, and the desired result is obtained by applying 4.5 to it. In the general case, choose a subgroup $\Gamma'' \subseteq \Gamma'$ of finite index such that Γ'' is normal in Γ ; this is possible because Γ' has only finitely many conjugates in Γ , so their intersection is still of finite index and is normal. Now compare $\varepsilon_{\Gamma''}$ to both $\varepsilon_{\Gamma'}$ and ε_Γ .]

5 Consequences of Swan's Theorem

In this section, finally, the reader will begin to see some Euler characteristics. We will then be ready in §6 to discuss Euler characteristics of *groups*.

For any finitely generated abelian group A , we define the *rank* of A (sometimes called the *torsion-free rank* of A) by

$$\text{rk}_\mathbb{Z}(A) = \dim_{\mathbb{Q}}(\mathbb{Q} \otimes_\mathbb{Z} A).$$

If A is free this obviously agrees with the naive rank [cardinality of a basis] which we have been using throughout this chapter for free modules; in general, $\text{rk}_{\mathbb{Z}}(A)$ as defined above is equal to the rank in this naive sense of the “free part” of A , i.e., of the quotient of A by its torsion subgroup. In particular, $\text{rk}_{\mathbb{Z}}(A) = 0$ if and only if A is finite.

Let C be a non-negative chain complex of abelian groups. We say that C is *finite dimensional* if $C_i = 0$ for sufficiently large i . If, in addition, each C_i is finitely generated, then C is called *finite*. Suppose that C is finite dimensional and that $H_* C$ is finitely generated; then we define the *Euler characteristic* $\chi(C)$ by

$$\chi(C) = \sum_{i \geq 0} (-1)^i \text{rk}_{\mathbb{Z}}(H_i C).$$

In case C is the cellular chain complex $C(X)$ of a finite dimensional CW-complex X , we write $\chi(X)$ instead of $\chi(C(X))$. Thus

$$\chi(X) = \sum_{i \geq 0} (-1)^i \text{rk}_{\mathbb{Z}}(H_i X).$$

If X is actually finite, then $\chi(X)$ is equal to the classical Euler characteristic $\sum_i (-1)^i n_i$, where n_i is the number of i -cells of X . This follows from:

(5.1) Proposition. *If C is a finite chain complex, then*

$$\chi(C) = \sum_i (-1)^i \text{rk}_{\mathbb{Z}}(C_i).$$

See, for instance, Spanier [1966], 4.3.14, or Dold [1972], V.5.2, for the (easy) proof.

It is also convenient sometimes to compute Euler characteristics using “mod p ” homology. This is justified by:

(5.2) Proposition. *Let C be a finite dimensional free chain complex over $\mathbb{Z}$ with $H_* C$ finitely generated. Let p be a prime number. Then*

$$\chi(C) = \sum_i (-1)^i \dim_{\mathbb{Z}_p}(H_i(C \otimes \mathbb{Z}_p)).$$

PROOF. Let $r_i = \dim_{\mathbb{Z}_p}((H_i C)_p)$ and $s_i = \dim_{\mathbb{Z}_p}(\rho(H_i C))$. [Recall that $A_p = A \otimes \mathbb{Z}_p = A/pA$ and that ${}_p A = \{a \in A : pa = 0\} = \text{Tor}(A, \mathbb{Z}_p)$ for any abelian group A .] Using the universal coefficient theorem (or the long exact homology sequence associated to the short exact sequence $0 \rightarrow C \xrightarrow{p} C \rightarrow C \otimes \mathbb{Z}_p \rightarrow 0$), one finds that $\dim_{\mathbb{Z}_p}(H_i(C \otimes \mathbb{Z}_p)) = r_i + s_{i-1}$. On the other hand, since $H_i C$ is a direct sum of cyclic groups, one sees easily that $r_i = \text{rk}_{\mathbb{Z}}(H_i C) + s_i$. Thus $\dim_{\mathbb{Z}_p}(H_i(C \otimes \mathbb{Z}_p)) = \text{rk}_{\mathbb{Z}}(H_i C) + s_i + s_{i-1}$, and the proposition follows at once. $\square$

We now prove the main result of this section:

(5.3) Theorem. *Let G be a finite group and let C be a finite dimensional chain complex of projective $\mathbb{Z}G$ -modules. If $H_*(C)$ is finitely generated then so is $H_*(C_G)$, and*

$$\chi(C) = |G| \cdot \chi(C_G).$$

(In other words, C behaves like a finite free complex as far as Euler characteristics are concerned.)

PROOF. In case C is a finite projective complex, this is immediate from 4.5, since χ can then be computed on the chain level (5.1). The general case is reduced to the finite case by means of Lemma 5.4 below. $\square$

Recall that a ring R is *noetherian* if every submodule of a finitely generated module is finitely generated. For example, $\mathbb{Z}G$ is noetherian if G is finite, since a finitely generated $\mathbb{Z}G$ -module is finitely generated as a $\mathbb{Z}$ -module.

(5.4) Lemma. *Let R be a noetherian ring and C a finite dimensional chain complex of projective R -modules. If H_*C is finitely generated over R , then C is homotopy equivalent to a finite projective complex.*

PROOF. First we construct, step-by-step, a free complex F of finite type which admits a weak equivalence $\tau: F \rightarrow C$. The inductive step is carried out as follows: Suppose we have constructed the n -skeleton of F together with a chain map

$$\begin{array}{ccccccc} F_n & \longrightarrow & \cdots & \longrightarrow & F_0 \\ \downarrow \tau & & & & \downarrow \tau \\ \cdots & \longrightarrow & C_{n+1} & \xrightarrow{\partial} & C_n & \longrightarrow & \cdots & \longrightarrow & C_0 \end{array}$$

such that $H_i\tau$ is an isomorphism for $i < n$ and an epimorphism for $i = n$. Then $\ker(H_n\tau)$ is finitely generated since F_n is finitely generated and R is noetherian. Let $x_1, \dots, x_r \in F_n$ be cycles representing generators of $\ker(H_n\tau)$, let $y_1, \dots, y_r \in C_{n+1}$ be chains such that $\partial y_i = \tau x_i$, and let $z_1, \dots, z_s \in C_{n+1}$ be cycles representing generators of the finitely generated module $H_{n+1}C$. Then we can take F_{n+1} to be free with basis $e_1, \dots, e_r, f_1, \dots, f_s$ and set $\partial e_i = x_i$, $\partial f_i = 0$, $\tau e_i = y_i$, and $\tau f_i = z_i$. The resulting chain map

$$\begin{array}{ccccccc} F_{n+1} & \longrightarrow & F_n & \longrightarrow & \cdots \\ \downarrow \tau & & \downarrow \tau & & \\ \cdots & \longrightarrow & C_{n+1} & \longrightarrow & C_n & \longrightarrow & \cdots \end{array}$$

induces an H_i -isomorphism for $i < n + 1$ and an H_{n+1} -epimorphism. This completes the inductive construction of F and τ . Note that the weak equivalence τ is actually a homotopy equivalence by I.8.4.

Now let n be an integer such that $C_i = 0$ for $i > n$, and consider the quotient $\bar{F}$ of F defined by

$$\bar{F}_i = \begin{cases} F_i & i < n \\ F_n/B_n & i = n \\ 0 & i > n, \end{cases}$$

B_n being the module of n -boundaries. Then we still have a weak equivalence $\bar{F} \rightarrow C$, and the lemma will clearly follow if we can show that F_n/B_n is projective. For any R -module M , we have

$$H^{n+1}(\mathcal{H}om_R(F, M)) \approx H^{n+1}(\mathcal{H}om_R(C, M)) = 0$$

since $F \simeq C$ and $C_{n+1} = 0$. Examining cocycles and coboundaries in $\mathcal{H}om_R(F, M)$ as in the proof of VIII.2.1, we conclude easily that B_n is a direct summand of F_n , and hence that F_n/B_n is projective. $\square$

To illustrate the significance of Theorem 5.3, we give two special cases.

(5.5) Corollary. *Let G be a finite group and let X be a finite dimensional free G -CW-complex with $H_* X$ finitely generated. Then $H_*(X/G)$ is finitely generated and $\chi(X) = |G| \cdot \chi(X/G)$.*

PROOF. Apply 5.3 to the cellular chain complex of X . $\square$

This result is obvious, of course, if X is finite, since we can then compute Euler characteristics by counting cells.

The second special case involves group cohomology. If Γ is a group such that $H_i \Gamma$ is finitely generated for all i and finite for sufficiently large i , then we set

$$\tilde{\chi}(\Gamma) = \sum_i (-1)^i \text{rk}_{\mathbb{Z}}(H_i \Gamma).$$

[The notation $\tilde{\chi}$ here is intended to suggest that this is not always the “right” Euler characteristic. We will explain this in §7.]

(5.6) Corollary. *Let Γ be a group with $\text{cd } \Gamma < \infty$ and let Γ' be a normal subgroup of finite index. If $H_* \Gamma'$ is finitely generated then so is $H_* \Gamma$, and $\tilde{\chi}(\Gamma') = (\Gamma : \Gamma') \cdot \tilde{\chi}(\Gamma)$.*

PROOF. Let P be a projective resolution of finite length of $\mathbb{Z}$ over $\mathbb{Z}\Gamma$, let $G = \Gamma/\Gamma'$, and let $C = P_{\Gamma'} = \mathbb{Z}G \otimes_{\mathbb{Z}\Gamma} P$ (cf. §II.2, exercise 3). Then C is a finite dimensional complex of projective $\mathbb{Z}G$ -modules with $H_* C = H_* \Gamma'$ and $H_*(C_G) = H_* \Gamma$; now apply 5.3. $\square$

Finally, we will need a mod p analogue of 5.3:

(5.7) Theorem. *Let G be a finite p -group for some prime p , let k be a field of characteristic p , and let C be a finite dimensional complex of projective kG -modules. If $H_* C$ is of finite rank over k , then so is $H_*(C_G)$, and $\chi(C) = |G| \cdot \chi(C_G)$, where $\chi(\quad)$ now denotes $\sum (-1)^i \dim_k(H_i(\quad))$.*

PROOF. This is identical to that of 5.3, with one major simplification; namely, the mod p analogue of 4.5 is now true for trivial reasons, since all projective kG -modules are free (VI.8.5). $\square$

One also has, obviously, mod p analogues of the corollaries 5.5 and 5.6 of 5.3.

EXERCISES

1. Let C and G be as in 5.3. For each $g \in G$, let $L(g)$ be the Lefschetz number of g acting on C , i.e.,

$$L(g) = \sum_i (-1)^i \operatorname{tr}_{\mathbb{Q}}(g \text{ acting on } H_i C \otimes \mathbb{Q}).$$

Prove that $L(g) = 0$ for $g \neq 1$, and state a topological corollary analogous to 5.5. [Hint: We may assume C is finite, so that $L(g)$ can be computed on the chain level. Exercise 3 of §4 now describes $L(g)$ in terms of Hattori-Stallings ranks, and the latter are computed by Swan's theorem (4.4).]

2. Let Γ and Γ' be as in 5.6. Let ψ_i be the character associated to the representation of the finite group Γ/Γ' acting on $H_i(\Gamma', \mathbb{Q})$. Show that $\sum_i (-1)^i \psi_i$ is an integral multiple of the character of the regular representation of Γ/Γ' . [The regular representation, by definition, is the free $\mathbb{Q}[\Gamma/\Gamma']$ -module of rank 1; its character takes the value 0 on all non-identity elements of Γ/Γ' .]

6 Euler Characteristics of Groups: The Torsion-Free Case

Euler characteristics of groups have been defined by a number of people under a number of different finiteness conditions. For our purposes the following condition is the most convenient one to impose: A group Γ is said to be of *finite homological type* if (i) $\operatorname{vcd} \Gamma < \infty$ and (ii) for every Γ -module M which is finitely generated as an abelian group, $H_i(\Gamma, M)$ is finitely generated for all i . [Warning: This definition differs slightly from that given in Brown [1974].] Note that if Γ is torsion-free, then (i) implies that $\operatorname{cd} \Gamma < \infty$ (cf. §VIII.3).

Remark. The main examples of groups of finite homological type are the groups of type *VFP*, and the reader may prefer to replace "finite homological

type” by “type *VFP*” in what follows; this results in occasional simplifications of the proofs. On the other hand, we will see that one cannot restrict attention to groups of type *VFL*, even though all known examples of groups of type *VFP* are in fact of type *VFL*.

(6.1) Lemma. *If Γ is a group and Γ' is a subgroup of finite index, then Γ is of finite homological type if and only if Γ' is of finite homological type.*

PROOF. It is obvious that $\text{vcd } \Gamma < \infty$ if and only if $\text{vcd } \Gamma' < \infty$, so we need only check that (ii) holds for Γ if and only if it holds for Γ' . The “only if” part follows from Shapiro’s lemma, since $\text{coind}_{\Gamma'}^{\Gamma} M$ is finitely generated over $\mathbb{Z}$ if M is. Conversely, suppose Γ' satisfies (ii). We know that Γ' has a subgroup Γ'' of finite index which is normal in Γ (cf. §4, hint to exercise 5), and Γ'' still satisfies (ii) by what we have just proved. Replacing Γ' by Γ'' , then, we may assume Γ' is normal in Γ . For any Γ -module M which is finitely generated over $\mathbb{Z}$, the Hochschild–Serre spectral sequence

$$E_{pq}^2 = H_p(\Gamma/\Gamma', H_q(\Gamma', M)) \Rightarrow H_{p+q}(\Gamma, M)$$

now shows that $H_i(\Gamma, M)$ is finitely generated for all i , as required. $\square$

Suppose now that Γ is of finite homological type and *torsion-free*. We then define the *Euler characteristic* $\chi(\Gamma)$ by

$$\chi(\Gamma) = \sum_i (-1)^i \text{rk}_{\mathbb{Z}}(H_i \Gamma).$$

[Thus $\chi(\Gamma) = \tilde{\chi}(\Gamma)$ in this case, where $\tilde{\chi}$ is as in §5.] In case Γ is of type *FP*, one has an analogue of the combinatorial formula for the Euler characteristic of a finite *CW*-complex. Namely, if P is a finite projective resolution of $\mathbb{Z}$ over $\mathbb{Z}$, then

$$(6.2) \quad \chi(\Gamma) = \sum_i (-1)^i \varepsilon(P_i),$$

where ε is the “rank” defined in §1. [To prove this, we need only note that $H_* \Gamma$ can be computed from the finite chain complex P_{Γ} ; the assertion now follows from 5.1.] We also have, trivially, the topological interpretation $\chi(\Gamma) = \chi(K(\Gamma, 1))$.

Here are a few examples where $\chi(\Gamma)$ can easily be computed.

EXAMPLES

1. If Γ is a free group of rank $n \geq 0$, then there is a $K(\Gamma, 1)$ with one vertex and n 1-cells, hence $\chi(\Gamma) = 1 - n$.

2. If Γ is a free abelian group of rank $n \geq 1$, then the n -torus $S^1 \times \cdots \times S^1$ is a $K(\Gamma, 1)$ with Euler characteristic $\chi(S^1) \cdots \chi(S^1) = 0$, hence $\chi(\Gamma) = 0$.

3. Let Γ be the commutator subgroup of $SL_2(\mathbb{Z})$. The latter is well-known to be an amalgamation $\mathbb{Z}_4 *_{\mathbb{Z}_2} \mathbb{Z}_6$. (See Serre [1977a], I.4.2, for an indication of an easy proof of this; see also example 3 of §VIII.9 above.) In particular, it follows that $SL_2(\mathbb{Z})_{ab}$ is of order $4 \cdot 6/2 = 12$, so that Γ is of index 12 in $SL_2(\mathbb{Z})$. One can also check, using the normal form for words in an amalgamated free product, that Γ is free of rank 2 (cf. Serre [1977a], I.1.3, proof of Prop. 4). Hence $\chi(\Gamma) = 1 - 2 = -1$ by example 1.

We will see more examples later.

The following property of $\chi(\)$ is fundamental:

(6.3) Theorem. *If Γ is torsion-free and of finite homological type and Γ' is a subgroup of finite index, then*

$$\chi(\Gamma') = (\Gamma : \Gamma') \cdot \chi(\Gamma).$$

PROOF. It suffices to prove this when Γ' is normal in Γ , in which case it follows from Corollary 5.6. $\square$

The proof of 6.3 simplifies slightly if Γ is of type *FP*; in this case we can use 6.2 and exercise 5 of §4 instead of 5.6. And the proof simplifies drastically if Γ is of type *FL*, since exercise 5 of §4 is trivial for free modules.

Theorem 6.3 has an interesting application to the theory of group extensions:

(6.4) Corollary. *Let $1 \rightarrow \Gamma \rightarrow E \rightarrow G \rightarrow 1$ be a group extension such that Γ is torsion-free and of finite homological type and G is of prime order p . If $p \nmid \chi(\Gamma)$, then the extension splits.*

(If Γ is free on n generators, for instance, then such an extension must split whenever $p \nmid (n - 1)$.)

PROOF. The group E necessarily has torsion; for if it were torsion-free, then 6.3 would be applicable and would yield $p \mid \chi(\Gamma)$, contrary to our hypothesis. Let $\tilde{G}$ be a non-trivial finite subgroup of E . Since Γ is torsion-free, $\tilde{G} \cap \Gamma = \{1\}$; thus $\tilde{G}$ maps injectively to G . But $|G|$ is prime, so $\tilde{G}$ maps isomorphically and provides a splitting. $\square$

Remarks

1. Even if we are interested in 6.4 only for groups Γ of type *FL*, the proof still requires a theory of Euler characteristics for groups of type *FP*. For we need to apply 6.3 to E in the first part of the proof (assuming E is torsion-free); and all we know about E , given that Γ is of type *FL*, is that it is of type *FP* if it is torsion-free.

2. We will give a substantial improvement of 6.4 later (cf. 9.4).

7 Extension to Groups with Torsion

From the topological point of view, it may seem very reasonable to restrict attention to torsion-free groups. Indeed, topologists usually consider Euler characteristics only for finite dimensional complexes, and we know that Γ must be torsion-free if there is a finite dimensional $K(\Gamma, 1)$. From the algebraic point of view, however, this seems less reasonable; for it forces us to exclude certain groups Γ (like $SL_2(\mathbb{Z})$) just because they have torsion, even though Γ may have subgroups of finite index for which χ is defined.

There is a simple solution to this problem, based on an idea due to Wall [1961]. Namely, if Γ is an arbitrary group of finite homological type, then we choose a torsion-free subgroup Γ' of finite index and set

$$\chi(\Gamma) = \frac{\chi(\Gamma')}{(\Gamma : \Gamma')} \in \mathbb{Q}.$$

This is motivated by 6.3, and, in fact, 6.3 is precisely what is needed to justify this definition, i.e., to show that the right-hand side is independent of the choice of Γ' . For suppose Γ'' is another such subgroup and let $\Gamma_0 = \Gamma' \cap \Gamma''$; then

$$\frac{\chi(\Gamma')}{(\Gamma : \Gamma')} \stackrel{(6.3)}{=} \frac{\chi(\Gamma_0)/(\Gamma' : \Gamma_0)}{(\Gamma : \Gamma')} = \frac{\chi(\Gamma_0)}{(\Gamma : \Gamma_0)}$$

and similarly $\chi(\Gamma'')/(\Gamma : \Gamma'') = \chi(\Gamma_0)/(\Gamma : \Gamma_0)$, which proves our assertion.

If Γ is finite, for instance, then we can take $\Gamma' = \{1\}$ and we find

$$(7.1) \quad \chi(\Gamma) = \frac{1}{|\Gamma|}.$$

Or if $\Gamma = SL_2(\mathbb{Z})$, then we can take Γ' to be the commutator subgroup (cf. example 3 of §6), and we find

$$(7.2) \quad \chi(SL_2(\mathbb{Z})) = -\frac{1}{12}.$$

These examples show that the rational number $\chi(\Gamma)$ is not generally an integer. In particular, $\chi(\Gamma)$ need not be equal to the naive Euler characteristic $\tilde{\chi}(\Gamma)$ defined in §5, which is always an integer. [Note that $\tilde{\chi}(\Gamma)$ is indeed defined if Γ is of finite homological type. For we know $H_i \Gamma$ is finitely generated for all i , and a transfer argument (III.10.1) shows that $\text{rk}_{\mathbb{Z}}(H_i \Gamma) = 0$ for $i > \text{vcd } \Gamma$.] This failure of $\chi(\Gamma)$ to be integral raises a number of interesting questions, which we will take up in §9.

In our study of Euler characteristics, we will attempt to get information about $\chi(\Gamma)$ by studying actions of Γ on CW -complexes. For this purpose, the *equivariant Euler characteristic* $\chi_{\Gamma}(X)$ is a fundamental tool. It is defined as follows: Suppose X is a Γ -complex such that (i) every isotropy group Γ_x

is of finite homological type and (ii) X has only finitely many cells mod Γ ; then we set

$$\chi_{\Gamma}(X) = \sum_{\sigma \in \mathcal{E}} (-1)^{\dim \sigma} \chi(\Gamma_{\sigma}),$$

where $\mathcal{E}$ is a set of representatives for the cells of X mod Γ . Note that $\chi(\Gamma) = \chi_{\Gamma}(\text{pt.})$, so the equivariant Euler characteristic can be viewed as a generalization of the ordinary Euler characteristic.

We now close this section by listing some useful properties of $\chi(\quad)$ and $\chi_{\Gamma}(\quad)$.

(7.3) Proposition. (a) *If Γ is torsion-free and of finite homological type then*

$$\chi(\Gamma) = \tilde{\chi}(\Gamma) = \sum_i (-1)^i \dim_{\mathbb{Z}_p} (H_i(\Gamma, \mathbb{Z}_p))$$

for any prime p .

(b) *If Γ is of finite homological type and Γ' is a subgroup of finite index, then*

$$\chi(\Gamma') = (\Gamma : \Gamma') \cdot \chi(\Gamma).$$

(b') *More generally, if $\chi_{\Gamma}(X)$ is defined and $\Gamma' \subseteq \Gamma$ is a subgroup of finite index, then $\chi_{\Gamma'}(X)$ is defined and*

$$\chi_{\Gamma'}(X) = (\Gamma : \Gamma') \cdot \chi_{\Gamma}(X).$$

(c) *Suppose that $\chi_{\Gamma}(X)$ is defined and that each Γ_{σ} is torsion-free. Then the equivariant homology $H_{\star}^{\Gamma}(X)$ is finitely generated and*

$$\chi_{\Gamma}(X) = \tilde{\chi}_{\Gamma}(X),$$

where $\tilde{\chi}_{\Gamma}(X) = \sum_i (-1)^i \text{rk}_{\mathbb{Z}}(H_i^{\Gamma}(X))$.

(d) *Let $1 \rightarrow \Gamma' \rightarrow \Gamma \rightarrow \Gamma'' \rightarrow 1$ be a short exact sequence of groups with Γ' and Γ'' of finite homological type. If Γ is virtually torsion-free, then Γ is of finite homological type and*

$$\chi(\Gamma) = \chi(\Gamma') \cdot \chi(\Gamma'').$$

(e) *Let $\Gamma = \Gamma_1 *_A \Gamma_2$, where Γ_1 , Γ_2 , and A are of finite homological type. If Γ is virtually torsion-free then Γ is of finite homological type and*

$$\chi(\Gamma) = \chi(\Gamma_1) + \chi(\Gamma_2) - \chi(A).$$

(e') *More generally, suppose X is a contractible Γ -complex such that $\chi_{\Gamma}(X)$ is defined. If Γ is virtually torsion-free, then Γ is of finite homological type and*

$$\chi(\Gamma) = \chi_{\Gamma}(X).$$

PROOF. The first equality of (a) is true by definition, and the second follows from 5.2. (b) is immediate from the definition of χ . To prove (b'), fix a cell σ of X and consider the set $\Gamma\sigma$ of cells which are equivalent to σ mod Γ . Since $\Gamma\sigma$ is in 1-1 correspondence with Γ/Γ_{σ} , it decomposes into finitely many

Γ' -orbits, represented by the cells $\gamma\sigma$ where γ ranges over a set S of representatives for $\Gamma'\backslash\Gamma/\Gamma_\sigma$. Note that $\Gamma'_{\gamma\sigma} = \Gamma' \cap \Gamma_{\gamma\sigma} = \Gamma' \cap \gamma\Gamma_\sigma\gamma^{-1}$, which is conjugate in Γ to $\gamma^{-1}\Gamma'\gamma \cap \Gamma_\sigma$. The contribution of $\Gamma\sigma$ to the sum defining $\chi_\Gamma(X)$ is therefore

$$\begin{aligned} \sum_{\gamma \in S} (-1)^{\dim \gamma\sigma} \chi(\Gamma' \cap \gamma\Gamma_\sigma\gamma^{-1}) &= (-1)^{\dim \sigma} \sum_{\gamma \in S} \chi(\gamma^{-1}\Gamma'\gamma \cap \Gamma_\sigma) \\ &= (-1)^{\dim \sigma} \sum_{\gamma \in S} (\Gamma_\sigma : \gamma^{-1}\Gamma'\gamma \cap \Gamma_\sigma) \cdot \chi(\Gamma_\sigma) \\ &= (-1)^{\dim \sigma} (\Gamma : \Gamma') \cdot \chi(\Gamma_\sigma). \end{aligned}$$

(The last equality here is obtained by decomposing $\Gamma'\backslash\Gamma$ into orbits under the right action of Γ_σ .) (b') now follows at once.

To prove (c), consider the equivariant homology spectral sequence (VII.7.7)

$$E_{pq}^1 = \bigoplus_{\sigma \in \mathcal{E}_p} H_q(\Gamma_\sigma, \mathbb{Z}_\sigma) \Rightarrow H_{p+q}^\Gamma(X),$$

where $\mathcal{E}_p$ is a set of representatives for the p -cells of X mod Γ . Since Γ_σ is torsion-free and of finite homological type, we know that $H_*(\Gamma_\sigma)$ is finitely generated. Hence $H_*^\Gamma(X)$ is finitely generated, and we can compute $\tilde{\chi}_\Gamma(X)$ from the E^1 -term of the spectral sequence (cf. VII.2.7). Assuming for the moment that Γ_σ acts trivially on $\mathbb{Z}_\sigma$ for all σ , we find

$$\begin{aligned} \tilde{\chi}_\Gamma(X) &= \sum_{p,q} \sum_{\sigma \in \mathcal{E}_p} (-1)^{p+q} \operatorname{rk}_{\mathbb{Z}}(H_q(\Gamma_\sigma)) \\ &= \sum_p \sum_{\sigma \in \mathcal{E}_p} (-1)^p \chi(\Gamma_\sigma) \\ &= \chi_\Gamma(X). \end{aligned}$$

In the general case, we need only note that all Euler characteristics can be computed from homology with $\mathbb{Z}_2$ -coefficients (cf. (a) above). Since Γ_σ acts trivially on $(\mathbb{Z}_2)_\sigma$, the result follows as above from the equivariant homology spectral sequence with $\mathbb{Z}_2$ -coefficients.

(d) Let $\Gamma_0 \subseteq \Gamma$ be a torsion-free subgroup of finite index whose image Γ_0'' in Γ'' is torsion-free, and let $\Gamma'_0 = \Gamma_0 \cap \Gamma'$. I claim that $(\Gamma : \Gamma_0) = (\Gamma' : \Gamma'_0) \cdot (\Gamma'' : \Gamma_0'')$; for $(\Gamma : \Gamma_0) = (\Gamma : \Gamma'\Gamma_0) \cdot (\Gamma'\Gamma_0 : \Gamma_0)$, and the well-known isomorphism laws of group theory imply that $(\Gamma : \Gamma'\Gamma_0) = (\Gamma'' : \Gamma_0'')$ and $(\Gamma'\Gamma_0 : \Gamma_0) = (\Gamma' : \Gamma'_0)$. We may therefore replace the given exact sequence by $1 \rightarrow \Gamma'_0 \rightarrow \Gamma_0 \rightarrow \Gamma_0'' \rightarrow 1$, i.e., we may assume that Γ' , Γ , and Γ'' are torsion-free. Then $\operatorname{cd} \Gamma < \infty$ by VIII.2.4b, and the Hochschild–Serre spectral sequence

$$E_{pq}^2 = H_p(\Gamma'', H_q(\Gamma', M)) \Rightarrow H_{p+q}(\Gamma, M)$$

shows that Γ is of finite homological type. Now take $M = \mathbb{Z}_2$. Since $H_*(\Gamma', \mathbb{Z}_2)$ is finite, there is a subgroup $\Gamma_0'' \subseteq \Gamma''$ of finite index which acts trivially on it. Replacing Γ'' by Γ_0'' and Γ by the inverse image of Γ_0'' , we may assume that Γ''

acts trivially on $H_*(\Gamma', \mathbb{Z}_2)$. Then $E_{pq}^2 = H_p(\Gamma'', \mathbb{Z}_2) \otimes_{\mathbb{Z}_2} H_q(\Gamma', \mathbb{Z}_2)$. Computing Euler characteristics from this spectral sequence, we conclude that $\chi(\Gamma) = \chi(\Gamma') \cdot \chi(\Gamma'')$.

Finally, (e) follows from (e') applied to the tree associated to $\Gamma_1 *_A \Gamma_2$ (cf. appendix to Chapter II), so it remains only to prove (e'). In view of (b'), it suffices to prove (e') when Γ is torsion-free. Since X is contractible, we have $H_*(\Gamma, M) \approx H_*^\Gamma(X, M)$ for any Γ -module M (VII.7.3), and similarly for cohomology. A spectral sequence argument (cf. proof of (c)) then shows that $H^i(\Gamma, M) = 0$ for $i > \dim X + \max\{\text{cd } \Gamma_\sigma\}$ and that $H_*(\Gamma, M)$ is finitely generated if M is. Hence Γ is of finite homological type. Moreover,

$$\begin{aligned}\chi(\Gamma) &= \tilde{\chi}(\Gamma) && \text{because } \Gamma \text{ is torsion-free} \\ &= \tilde{\chi}_\Gamma(X) && \text{because } H_* \Gamma \approx H_*^\Gamma(X) \\ &= \chi_\Gamma(X) && \text{by (c).}\end{aligned}\quad \square$$

As an example of (e), consider $SL_2(\mathbb{Z}) \approx \mathbb{Z}_4 *_2 \mathbb{Z}_6$. Then (e) gives

$$\chi(SL_2(\mathbb{Z})) = \frac{1}{4} + \frac{1}{6} - \frac{1}{2} = -\frac{1}{12},$$

in agreement with 7.2.

EXERCISES

1. The purpose of this exercise is to outline an alternative method of defining a "rational Euler characteristic" under suitable finiteness hypotheses. This is due to Bass [1976], Chiswell [1976b] and Stallings [unpublished]. For any group Γ and any finitely generated projective $\mathbb{Q}\Gamma$ -module P , we can define a "rational rank" $\rho(P)$ [or $\rho_\Gamma(P)$] by $\rho(P) = R_{\mathbb{Q}\Gamma}(P)(1) \in \mathbb{Q}$. [This is motivated by 4.2.] Note that $\rho_{\Gamma'}(P) = (\Gamma : \Gamma')\rho_\Gamma(P)$ if $\Gamma' \subseteq \Gamma$ is a subgroup of finite index (cf. 4.1).

(a) Γ is said to be of type FP over $\mathbb{Q}$ if $\mathbb{Q}$ admits a finite projective resolution over $\mathbb{Q}\Gamma$. In this case we choose such a resolution P and set $e(\Gamma) = \sum (-1)^i \rho(P_i)$. Show that $e(\Gamma)$ is well-defined and that $e(\Gamma') = (\Gamma : \Gamma')e(\Gamma)$ if $\Gamma' \subseteq \Gamma$ is a subgroup of finite index.

(b) Show that any group of type VFP is of type FP over $\mathbb{Q}$. [Hint: If P is a $\mathbb{Q}\Gamma$ -module which is projective over $\mathbb{Q}\Gamma'$, where $(\Gamma : \Gamma') < \infty$, then P is projective. This can be proved by an averaging argument.]

(c) If Γ is of type VFL , show that $e(\Gamma) = \chi(\Gamma)$.

[**Remark.** It is not known whether $e(\Gamma) = \chi(\Gamma)$ for arbitrary groups Γ of type VFP .]

2. (a) If $e(\Gamma) \neq 0$, show that $C(\Gamma)$, the center of Γ , is finite. [Consider the "complete Euler characteristic" $E(\Gamma) \in T(\mathbb{Q}\Gamma)$, defined by $E(\Gamma) = \sum (-1)^i R_{\mathbb{Q}\Gamma}(P_i)$, where (P_i) is a finite projective resolution of $\mathbb{Q}$ over $\mathbb{Q}\Gamma$. Note that $E(\Gamma)$ is simply $\text{tr}_{\mathbb{Q}\Gamma}(\text{id}_{\mathbb{Q}})$, as defined in exercise 2 of §2, and that $E(\Gamma)(1) = e(\Gamma)$. If $\gamma \in C(\Gamma)$, then we find (using exercise 3 of §2) that $E(\Gamma) = \text{tr}(\text{id}_{\mathbb{Q}}) = \text{tr}(\gamma \cdot \text{id}_{\mathbb{Q}}) = \gamma \cdot \text{tr}(\text{id}_{\mathbb{Q}}) = \gamma \cdot E(\Gamma)$. Hence $e(\Gamma) = E(\Gamma)(\gamma^{-1})$.]

(b) Deduce the following theorem of Gottlieb [1965]: If Y is a finite $K(\Gamma, 1)$ -complex with $\chi(Y) \neq 0$, then $C(\Gamma) = \{1\}$.

Remark. This proof of Gottlieb's theorem is due to Stallings [1965b], as reformulated by Bass [1976].

- *3. Suppose Γ is torsion-free and of finite homological type and let M be a Γ -module which is finitely generated as an abelian group. Show that

$$\sum_i (-1)^i \operatorname{rk}_{\mathbb{Z}}(H_i(\Gamma, M)) = \chi(\Gamma) \cdot \operatorname{rk}_{\mathbb{Z}}(M).$$

- *4. If Γ is of finite homological type and M is a Γ -module which is finitely generated as an abelian group, show that $H^i(\Gamma, M)$ is finitely generated for all i . [Hint: Reduce to the case where M is $\mathbb{Z}$ -free. In this case let $M^* = \operatorname{Hom}(M, \mathbb{Z})$ and note that $\mathcal{H}om_{\Gamma}(P, M) \approx \mathcal{H}om(P \otimes_{\Gamma} M^*, \mathbb{Z})$.]
- *5. Let $\Gamma \subset SL_2(\mathbb{Z})$ be a torsion-free subgroup of finite index k . Prove that Γ is a free group on $1 + k/12$ generators. Example: The principal congruence subgroup of level 3 has index $24 = |SL_2(\mathbb{Z}_3)|$, hence it is free on 3 generators.

8 Euler Characteristics and Number Theory

The study of groups of integral matrices is intimately related to algebraic number theory. For example, if one tries to classify elements of order p in $GL_n(\mathbb{Z})$ up to conjugacy, one quickly finds that ideal classes of the p -th cyclotomic field come into play. Recall now that many integral matrix groups are of type VFL (cf. §VIII.11). We therefore have rational numbers $\chi(\Gamma)$ associated to such groups, and it is natural to ask whether these numbers have any number-theoretic significance. The remarkable answer is that $\chi(\Gamma)$, for many arithmetic groups Γ , can be expressed in terms of values of zeta-functions. In this section we will give a brief survey of results of this type. See Serre [1971, 1979] for more details and a guide to the literature.

The starting point for the computation of $\chi(\Gamma)$ is the Gauss-Bonnet theorem of differential geometry, which says the following: Suppose Y is a closed Riemannian manifold. Then there is a measure μ on Y , constructed canonically from the curvature associated to the Riemannian metric, such that

$$\chi(Y) = \mu(Y).$$

[Note: μ is not in general a *positive* measure. If Y is a surface, for example, then $\mu = K \cdot dA/2\pi$, where K is the Gaussian curvature function and dA is the canonical "area" measure on Y .] We call μ the *Gauss-Bonnet measure* on Y .

If Y is a $K(\Gamma, 1)$ and X is its universal cover, then we can rewrite this formula as

$$(8.1) \quad \chi(\Gamma) = \mu(X/\Gamma).$$

Moreover, we can interpret the right-hand side as the measure of a fundamental domain for the action of Γ on X , relative to the Gauss–Bonnet measure on X .

We now specialize to the arithmetic case. Suppose G is a semi-simple algebraic group defined over $\mathbb{Q}$ and Γ is a *torsion-free* arithmetic subgroup of $G(\mathbb{Q}) \subseteq G(\mathbb{R})$. Let K be a maximal compact subgroup of $G(\mathbb{R})$ and let $X = G(\mathbb{R})/K$. Since K is compact, it is easy to see that X admits a Riemannian metric invariant under the action of $G(\mathbb{R})$; hence there is a $G(\mathbb{R})$ -invariant Gauss–Bonnet measure μ on X , and 8.1 holds if Γ is co-compact in $G(\mathbb{R})$. Using the compactness of K again, we can lift μ to a left-invariant measure (still called μ) on $G(\mathbb{R})$; 8.1 then takes the form

$$(8.2) \quad \chi(\Gamma) = \mu(\Gamma \backslash G(\mathbb{R})).$$

The advantage of this formulation is that it remains valid even if Γ has torsion. For both sides of 8.2 get multiplied by $(\Gamma : \Gamma')$ if we replace Γ by a torsion-free subgroup Γ' of finite index.

We derived 8.2 under the assumption that Γ was co-compact in $G(\mathbb{R})$, and, as we saw in §VIII.9, this situation is far from typical. A remarkable theorem of Harder [1971], however, says that 8.2 remains valid even if Γ is not co-compact. Harder went on to explicitly calculate the measure μ for an interesting family of groups G , the so-called “Chevalley groups.” [There is one such group for every simple Lie algebra over $\mathbb{C}$. Thus we have the special linear groups SL_n , the symplectic groups Sp_{2n} , the special orthogonal groups SO_n , and five exceptional groups G_2, F_4, E_6, E_7 , and E_8 .]

For such G there is a canonical Haar measure (i.e., positive invariant measure) μ_a on $G(\mathbb{R})$, called the *arithmetic measure*, with the property that $\mu_a(G(\mathbb{R})/G(\mathbb{Z}))$ is expressible in terms of values of the zeta-function. On the other hand the Gauss–Bonnet measure μ , being invariant, must have the form $\mu = c\mu_a$ for some $c \in \mathbb{R}$. What Harder did was to calculate the scalar c , so that 8.2 would yield an explicit formula for $\chi(G(\mathbb{Z}))$ in terms of values of the zeta-function. We will give some examples of this formula below, following a brief review of the Riemann zeta-function.

Recall that $\zeta(s)$ is defined for $\operatorname{Re}(s) > 1$ by

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}}.$$

Thus $\zeta(s)$ is an analytic function in the half-plane $\operatorname{Re}(s) > 1$, and one extends it to a meromorphic function in the entire complex plane. Moreover, the function $\xi(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$ satisfies the functional equation $\xi(s) = \xi(1-s)$. We will be interested in the values of $\zeta(s)$ at negative integers. At *even* negative integers, this is trivial to compute:

$$\zeta(-2k) = 0 \quad \text{if } k \geq 1.$$

For $\xi(s)$ is holomorphic at $s = 2k + 1$, hence also at $s = 1 - (2k + 1) = -2k$. But $\Gamma(s/2)$ has a pole at $s = -2k$, so $\zeta(s)$ must have a zero there. At

odd negative integers, on the other hand, $\zeta(s)$ is non-zero and can be described in terms of the Bernoulli numbers, whose definition we now recall.

The function $z/(e^z - 1)$ has a power series expansion of the form

$$\frac{z}{e^z - 1} = 1 - \frac{z}{2} + \sum_{k=1}^{\infty} \frac{B_{2k}}{(2k)!} z^{2k},$$

where $B_{2k} \in \mathbb{Q}$. The numbers B_{2k} which arise in this way are called the *Bernoulli numbers*. Here are the first few of them, with their numerators and denominators factored into primes:

$$\begin{array}{lll} B_2 = \frac{1}{2 \cdot 3} & B_4 = -\frac{1}{2 \cdot 3 \cdot 5} & B_6 = \frac{1}{2 \cdot 3 \cdot 7} \\ B_8 = -\frac{1}{2 \cdot 3 \cdot 5} & B_{10} = \frac{5}{2 \cdot 3 \cdot 11} & B_{12} = -\frac{691}{2 \cdot 3 \cdot 5 \cdot 7 \cdot 13} \\ B_{14} = \frac{7}{2 \cdot 3} & B_{16} = -\frac{3617}{2 \cdot 3 \cdot 5 \cdot 17} & B_{18} = \frac{43867}{2 \cdot 3 \cdot 7 \cdot 19} \\ B_{20} = -\frac{283 \cdot 617}{2 \cdot 3 \cdot 5 \cdot 11} & B_{22} = \frac{11 \cdot 131 \cdot 593}{2 \cdot 3 \cdot 23} & B_{24} = -\frac{103 \cdot 2294797}{2 \cdot 3 \cdot 5 \cdot 7 \cdot 13} \end{array}$$

It is well-known that the Bernoulli numbers arise when one computes $\zeta(2k)$ for an integer $k \geq 1$:

$$\zeta(2k) = -\frac{(-1)^k 2^{2k-1} B_{2k}}{(2k)!} \pi^{2k}.$$

Using the functional equation to rewrite this as a formula for $\zeta(1 - 2k)$, we find that the factorial and the powers of -1 , 2 , and π disappear; the result is

$$(8.3) \quad \zeta(1 - 2k) = -\frac{B_{2k}}{2k}.$$

We can now state a few examples of Harder's formula for $\chi(G(\mathbb{Z}))$. First, taking $G = SL_2$, we have

$$(8.4) \quad \chi(SL_2(\mathbb{Z})) = \zeta(-1).$$

Since $\zeta(-1) = -B_2/2 = -\frac{1}{12}$, this agrees with 7.2. The group SL_2 is part of two infinite families of groups, namely, the special linear groups SL_n and the symplectic groups Sp_{2n} . [One has $Sp_2 = SL_2$.] The generalizations of 8.4 to these families are

$$(8.5) \quad \chi(SL_n(\mathbb{Z})) = \prod_{k=2}^n \zeta(1 - k)$$

and

$$(8.6) \quad \chi(Sp_{2n}(\mathbb{Z})) = \prod_{k=1}^n \zeta(1 - 2k).$$

Of course 8.5 involves the factor $\zeta(-2)$ as soon as $n \geq 3$, so it simply says $\chi(SL_n(\mathbb{Z})) = 0$ for $n \geq 3$. But 8.6 involves only non-zero values of $\zeta(s)$ and hence is more interesting.

For our last example we take G to be the exceptional Chevalley group E_7 . Harder's formula in this case turns out to involve ζ at $-1, -5, -7, -9, -11, -13$, and -17 , hence it involves B_k for $k = 2, 6, 8, 10, 12, 14$, and 18 . The precise formula turns out to be

$$(8.7) \quad \chi(E_7(\mathbb{Z})) = -\frac{691 \cdot 43867}{2^{21} \cdot 3^9 \cdot 5^2 \cdot 7^3 \cdot 11 \cdot 13 \cdot 19}.$$

Finally, we mention that Harder's formulas (such as 8.5 and 8.6) are valid with $\mathbb{Z}$ replaced by the ring of integers in a number field F and ζ replaced by the Dedekind ζ -function ζ_F associated to F . And Serre has shown that one can generalize even further, namely, to rings $\mathcal{O}_S$ of "S-integers". Here S is a finite set of primes of F and $\mathcal{O}_S \subset F$ is the subring consisting of the elements of F which are integral at all primes not in S . One then uses the ζ -function $\zeta_{F,S}$ obtained from ζ_F by omitting from its infinite product expansion the factors corresponding to the primes in S . If $F = \mathbb{Q}$ and S consists of a single prime p , for example, then $\zeta_{F,S}(s) = \zeta(s)(1 - p^{-s})$ and the generalization of 8.4 gives

$$\chi(SL_2(\mathbb{Z}[1/p])) = \zeta_{F,S}(-1) = \frac{p-1}{12}.$$

9 Integrality Properties of $\chi(\Gamma)$

Let Γ be a group of finite homological type (e.g., a group of type VFP). We know that $\chi(\Gamma) = \tilde{\chi}(\Gamma) \in \mathbb{Z}$ if Γ is torsion-free, so the non-integrality of $\chi(\Gamma)$ in general is somehow due to the presence of torsion in Γ . The rest of this chapter will be devoted to proving various precise versions of this vague statement.

The most obvious integrality statement that one can make in general is that $(\Gamma : \Gamma') \cdot \chi(\Gamma) \in \mathbb{Z}$ if Γ' is a torsion-free subgroup of finite index; for $(\Gamma : \Gamma') \cdot \chi(\Gamma) = \chi(\Gamma') = \tilde{\chi}(\Gamma') \in \mathbb{Z}$. Consequently,

$$(9.1) \quad d \cdot \chi(\Gamma) \in \mathbb{Z},$$

where $d = \gcd\{(\Gamma : \Gamma') : \Gamma' \text{ is torsion-free and of finite index}\}$. This simple observation, in conjunction with Harder's deep results (§8), already has non-trivial applications. For example, Serre [1971] used it to prove integrality

results about values of the zeta-function of a number field. Another application (also due to Serre) concerns the torsion in the exceptional Chevalley groups. It is based on:

(9.2) Lemma. *The prime divisors of d are precisely the primes p such that Γ has p -torsion.*

PROOF. Let H be a finite subgroup of Γ . If $\Gamma' \subseteq \Gamma$ is a torsion-free subgroup, then H acts freely on Γ/Γ' , so $(\Gamma:\Gamma')$, if finite, is divisible by $|H|$. Hence $|H|$ divides d . In particular, it follows that d is divisible by any prime p such that Γ has p -torsion. Conversely, suppose p is a prime such that Γ has no p -torsion. Let $\Gamma_0 \subseteq \Gamma$ be a torsion-free normal subgroup of finite index and let Γ' be an intermediate subgroup ($\Gamma_0 \subseteq \Gamma' \subseteq \Gamma$) such that Γ'/Γ_0 is a p -Sylow subgroup of Γ/Γ_0 . Then $(\Gamma':\Gamma_0)$ is a power of p and $(\Gamma:\Gamma')$ is relatively prime to p . By the first part of the proof, any finite subgroup of Γ' must be a p -group, so Γ' is torsion-free. But then $d \nmid (\Gamma:\Gamma')$, so $p \nmid d$. $\square$

Looking at formula 8.7 for $\chi(E_7(\mathbb{Z}))$, for example, we conclude from 9.1 and 9.2 that $E_7(\mathbb{Z})$ must have p -torsion for $p = 2, 3, 5, 7, 11, 13, 19$.

The proof of 9.2 shows that d is divisible by the order of every finite subgroup of Γ , hence d is divisible by the least common multiple m of the orders of the finite subgroups. Moreover, it is clear from 9.2 that d and m have the same prime divisors. But in general $d \neq m$. If $\Gamma = SL_3(\mathbb{Z})$, for instance, it can be shown that $d = 2^4 \cdot 3$ whereas $m = 2^3 \cdot 3$.

It is reasonable to ask, then, whether 9.1 can be improved to the statement $m \cdot \chi(\Gamma) \in \mathbb{Z}$. This question was raised by Serre, and it turns out to have an affirmative answer:

(9.3) Theorem (Brown [1974]). *Let Γ be a group of finite homological type and let m be the least common multiple of the orders of the finite subgroups of Γ . Then $m \cdot \chi(\Gamma) \in \mathbb{Z}$. Consequently, if a prime power p^a divides the denominator of $\chi(\Gamma)$, then Γ has a subgroup of order p^a .*

The second assertion follows easily from the first and the Sylow theorems. The proof of the first assertion requires some topological ideas involving finite group actions; it will be given in the next section.

Using 9.3 one can improve Serre's integrality results on the values of the zeta function (cf. Brown [1974], §9). One can also obtain good estimates of the amount of torsion in some of the exceptional Chevalley groups. For instance, 9.3 and 8.7 show that $E_7(\mathbb{Z})$ has subgroups of order 2^{21} , 3^9 , etc. See Serre [1979] for more information on this. A third application is the following improvement of 6.4:

(9.4) Corollary. *Let $1 \rightarrow \Gamma \rightarrow E \rightarrow G \rightarrow 1$ be a group extension such that Γ is torsion-free and of finite homological type and G is a p -group for some prime p . If $p \nmid \chi(\Gamma)$, then the extension splits.*

PROOF. We have $\chi(E) = \chi(\Gamma)/|G|$, and this fraction is in lowest terms since $p \nmid \chi(\Gamma)$. Theorem 9.3 therefore implies that E has a finite subgroup $\tilde{G}$ with $|\tilde{G}| = |G|$. Since Γ is torsion-free, $\tilde{G}$ maps isomorphically to G and provides a splitting. $\square$

10 Proof of Theorem 9.3; Finite Group Actions

We begin by interpreting Theorem 9.3 topologically. Since $\text{vcd } \Gamma < \infty$, we know from VIII.11.1 that there is a finite dimensional contractible Γ -complex X with finite isotropy groups. Moreover, I claim that X can be taken to be an *admissible* Γ -complex, in the sense that the following condition holds: For each cell σ of X , the isotropy group Γ_σ fixes σ pointwise. To see this, we need only recall that X can be taken to be simplicial (VIII.11.2); passing to the barycentric subdivision if necessary, we can assume that X is an *ordered* simplicial complex and that Γ is order-preserving. Then clearly no element of Γ can permute the vertices of a simplex non-trivially, so the admissibility condition holds.

Let $\Gamma' \subseteq \Gamma$ be a torsion-free normal subgroup of finite index. Then Γ' acts freely on X and the quotient $Y = X/\Gamma'$ is a $K(\Gamma', 1)$. Thus

$$\chi(\Gamma) = \frac{\tilde{\chi}(\Gamma')}{(\Gamma: \Gamma')} = \frac{\chi(Y)}{(\Gamma: \Gamma')},$$

where $\chi(Y) = \sum (-1)^i \text{rk}_{\mathbb{Z}}(H_i(Y))$. What we are trying to prove, then, is that

$$\frac{m}{(\Gamma: \Gamma')} \cdot \chi(Y) \in \mathbb{Z},$$

where $m = \text{lcm}\{|H|: H \text{ is a finite subgroup of } \Gamma\}$. In other words, we want to prove that

$$k \mid \chi(Y),$$

where k is the integer $(\Gamma: \Gamma')/m$.

Let G be the finite quotient Γ/Γ' . Note that the action of Γ on X induces an action of G on Y . Moreover, I claim that the isotropy group G_τ of any cell τ of Y is simply $\pi(\Gamma_\sigma)$, where σ is any cell of X lying over τ and $\pi: \Gamma \rightarrow G$ is the quotient map. For suppose $g\tau = \tau$ and choose $\gamma \in \Gamma$ such that $\pi(\gamma) = g$. Then $\gamma\sigma$ and σ both lie over τ , so $\gamma'\gamma\sigma = \sigma$ for some $\gamma' \in \Gamma'$. Since $\pi(\gamma'\gamma) = g$, this shows that $G_\tau \subseteq \pi(\Gamma_\sigma)$. The opposite inclusion is trivially true, whence the claim. (Incidentally, π actually maps Γ_σ isomorphically onto G_τ ; for $\Gamma_\sigma \cap \Gamma'$ is finite and torsion-free, hence trivial.)

Two facts follow from this. First, the action of G on Y is admissible. Second, every isotropy group G_τ has order dividing m , hence the cardinality $|G|/|G_\tau|$ of the orbit of τ is divisible by $k = |G|/m$.

Theorem 9.3 now follows from:

(10.1) Theorem. *Let G be a finite group and let Y be a finite dimensional admissible G -complex such that $H_* Y$ is finitely generated. If k is an integer which divides the cardinality of every G -orbit, then $k \mid \chi(Y)$.*

Note that it does not matter whether we interpret G -orbit to mean G -orbit of cells or G -orbit of points; for the admissibility condition implies that $G_\tau = G_y$ for any interior point y of τ .

Note also that the theorem is trivially true if Y is finite, since $\chi(Y)$ can then be computed by counting cells. The point of 10.1 is that it is true under fairly mild *homological* finiteness hypotheses on Y . In fact, one can even prove a version of 10.1 for spaces Y which are not CW -complexes (cf. Brown [1979], Theorem 7.4), but the proof then requires more machinery than is needed for 10.1.

PROOF OF 10.1. For any subgroup $H \subseteq G$, let $Y_H = \{y \in Y: G_y = H\}$. The admissibility condition implies that Y_H is a union of (open) cells, namely, those cells τ such that $G_\tau = H$. Note that $g \cdot Y_H = Y_{gHg^{-1}}$ for any $g \in G$. In particular, Y_H is stable under the action of the normalizer $N(H)$ of H in G . Moreover, $N(H)/H$ acts freely on Y_H . We can now easily explain the idea of the proof. Y is the disjoint union (set-theoretically) of the subspaces Y_H , so one might expect

$$(*) \quad \chi(Y) = \sum_{H \subseteq G} \chi(Y_H).$$

Now $Y_{gHg^{-1}} = g \cdot Y_H \approx Y_H$, so we can group together the terms in $(*)$ corresponding to conjugate subgroups H , and we obtain

$$(**) \quad \chi(Y) = \sum_{H \in \mathcal{C}} (G: N(H)) \chi(Y_H),$$

where $\mathcal{C}$ is a set of representatives for the conjugacy classes of subgroups of G which occur as isotropy groups in Y . [We have used the elementary fact that $(G: N(H))$ is the number of conjugates of H in G .] Since $N(H)/H$ acts freely on Y_H , 5.5 should now imply that $\chi(Y_H)$ is divisible by $(N(H): H)$, so that $(G: N(H))\chi(Y_H)$ is divisible by $(G: H)$. But $(G: H)$ is the cardinality of an orbit, so every term on the right-hand side of $(**)$ is divisible by k ; hence $k \mid \chi(Y)$.

We now must clarify and justify the steps in this "proof." In the first place, Y_H is not in general a subcomplex of Y , since it need not be closed. (A boundary point of a cell can have a bigger isotropy group than the interior points.) But each fixed-point set Y^H is a subcomplex, and hence Y_H is a difference of subcomplexes:

$$Y_H = Y^H - Y^{>H},$$

where $Y^{>H} = \bigcup_{H' \not\supseteq H} Y^{H'}$. Let's *assume* temporarily that each Y^H has finitely generated homology. Then the same is true of $Y^{>H}$ by a Mayer-Vietoris argument, since the family $\{Y^{H'}: H' \not\supseteq H\}$ is closed under intersection. Hence $H_*(Y^H, Y^{>H})$ is finitely generated, and I claim that (*) becomes true if we replace $\chi(Y_H)$ by $\chi(Y^H, Y^{>H})$.

To prove this, let $H_1, \dots, H_n$ be the subgroups of G , ordered so that $|H_i| \geq |H_{i+1}|$. We filter Y by subcomplexes

$$\emptyset = Y_0 \subseteq Y_1 \subseteq \dots \subseteq Y_n = Y,$$

defined inductively by

$$Y_i = Y_{i-1} \cup Y^{H_i}.$$

Note that $Y_{i-1} \cap Y^{H_i} = Y^{>H_i}$. We therefore have an excision isomorphism

$$H_*(Y_i, Y_{i-1}) \approx H_*(Y^{H_i}, Y^{>H_i}).$$

[Note that, because we are dealing with subcomplexes, there is no problem justifying the excision. Indeed, we already have an excision isomorphism on the level of cellular chain complexes.] Hence

$$\begin{aligned} \chi(Y) &= \sum_{i=1}^n \chi(Y_i, Y_{i-1}) \\ &= \sum_{i=1}^n \chi(Y^{H_i}, Y^{>H_i}), \end{aligned}$$

as claimed.

The remainder of our initial heuristic proof is now valid, with Y_H replaced everywhere by $(Y^H, Y^{>H})$. One needs, of course, to apply a relative version of 5.5 to $N(H)/H$ acting on $(Y^H, Y^{>H})$, but this presents no problem—simply apply 5.3 to the cellular chain complex $C(Y^H, Y^{>H})$, which is a finite dimensional complex of free $\mathbb{Z}[N(H)/H]$ -modules.

If we now drop the assumption that $H_*(Y^H)$ is finitely generated for each H , then we can argue as follows. Suppose first that G is a p -group for some prime p . Since $H_* Y$ is finitely generated, we can compute $\chi(Y)$ from $H_*(Y; \mathbb{Z}_p)$. We now repeat the argument above using $H_*(\quad; \mathbb{Z}_p)$ and using 5.7 instead of 5.3. This time, however, we have a *theorem* (VII.10.5 and exercise 2 of §VII.10) which says that $H_*(Y^H; \mathbb{Z}_p)$ is finitely generated, so we do not need to assume this.

Finally, we can easily reduce the general case to the p -group case: To prove 10.1, it suffices to show that if p^a is a prime power which divides k then $p^a | \chi(Y)$. Let $G(p)$ be a p -Sylow subgroup of G . By hypothesis $p^a | (G: G_\tau) = |G|/|G_\tau|$ for any cell τ of Y . Looking at the p -part of $|G|$ and $|G_\tau|$, we conclude that p^a divides $|G(p)|/|P|$, where P is any p -subgroup of G_τ . In particular, p^a divides $|G(p)|/|G(p)_\tau|$, which is the cardinality of a typical $G(p)$ -orbit. It now follows from the p -group case that $p^a | \chi(Y)$. $\square$

EXERCISE

Use the method of proof of 10.1 to prove the following “Lefschetz fixed-point theorem”: Let G be a finite group and Y an admissible finite dimensional G -complex. Assume that each fixed-point set Y^g ($g \in G$) has finitely generated homology. Prove that the Lefschetz number $L(g)$ is given by

$$L(g) = \chi(Y^g).$$

[Hint: You may assume G is cyclic, generated by g . Then g is non-trivial in $N(H)/H = G/H$ for every $H \subsetneq G$, so $L(g|Y_H) = 0$ by exercise 1 of §5. Now use the Lefschetz number analogue of (*) and note that every term is zero except the one corresponding to $H = G$.]

Remark. One can use this theorem to prove the following theorem about Euler characteristics of groups (cf. Brown [1982]): Let Γ be a group such that the centralizer $C(\gamma)$ is of finite homological type for every $\gamma \in \Gamma$ of finite order (including $\gamma = 1$). Let $\mathcal{C}$ be a set of representatives for the conjugacy classes of elements of finite order. Then $\mathcal{C}$ is finite, and

$$\tilde{\chi}(\Gamma) = \sum_{\gamma \in \mathcal{C}} \chi(C(\gamma)).$$

Consequently, we have the following formula, which expresses the deviation between $\chi(\Gamma)$ and $\tilde{\chi}(\Gamma)$ in terms of the torsion in Γ :

$$\tilde{\chi}(\Gamma) = \chi(\Gamma) + \sum_{\gamma \in \mathcal{C}'} \chi(C(\gamma)),$$

where $\mathcal{C}' = \mathcal{C} - \{1\}$.

11 The Fractional Part of $\chi(\Gamma)$

Under suitable hypotheses on Γ it is possible to deduce from the proof of 9.3 precise formulas expressing $\chi(\Gamma) - \tilde{\chi}(\Gamma)$ in terms of the torsion in Γ . Since $\tilde{\chi}(\Gamma) \in \mathbb{Z}$, such a formula can be regarded as a formula for the “fractional part” of the rational number $\chi(\Gamma)$. One formula of this type was just written down, following the exercise of §10. Another formula, which can be found in Brown [1974], has the form

$$\chi(\Gamma) = \tilde{\chi}(\Gamma) + \sum_H \frac{c(H)}{|H|},$$

where H ranges over the non-trivial finite subgroups of Γ (up to conjugacy) and $c(H)$ is an integer whose definition is too complicated to repeat here.

Rather than state and prove this formula, we will use similar methods to prove some formulas which are less precise but easier to prove and apply.

If S is a partially ordered set, then there is an obvious way to construct from S an ordered simplicial complex $|S|$. Namely, the vertices of $|S|$ are

the elements of S , and the simplices of $|S|$ are the linearly ordered finite subsets $s_0 < \cdots < s_n$ of S . [We have already seen one example of this construction: If K is any simplicial complex and S is the set of simplices of K , ordered by the face relation, then $|S|$ is simply the barycentric subdivision of K , cf. §VIII.11.] If Γ acts on S (and preserves the ordering), then there is an induced action of Γ on $|S|$, and we set

$$\chi_r(S) = \chi_r(|S|)$$

if the equivariant Euler characteristic on the right-hand side is defined. More generally, we will use the functor $S \mapsto |S|$ to assign arbitrary topological concepts to partially ordered sets. For instance we will write $C_*(S)$ and $H_*(S)$ instead of $C_*(|S|)$ and $H_*(|S|)$, and we will say that S is contractible or acyclic if $|S|$ is.

(11.1) Theorem. *Let Γ be a group of finite homological type. Assume that Γ has only finitely many finite subgroups (up to conjugacy) and that the normalizer $N(H)$ of each finite subgroup H has finite homological type. Let S be the set of non-trivial finite subgroups of Γ , regarded as a partially ordered set under inclusion, with Γ -action by conjugation. Then $\chi_r(S)$ is defined and*

$$\chi(\Gamma) \equiv \chi_r(S) \pmod{\mathbb{Z}}.$$

(In concrete terms, this gives the fractional part of $\chi(\Gamma)$ in terms of the Euler characteristics of subgroups of Γ of the form $N(H_0) \cap \cdots \cap N(H_n)$, where $H_0 \subset \cdots \subset H_n$ is a chain of non-trivial finite subgroups.)

PROOF. To show $\chi_r(S)$ is defined, we must show that $|S|$ has only finitely many simplices $\sigma \bmod \Gamma$ and that each Γ_σ has finite homological type. Let $\mathcal{F}$ be a (finite) set of representatives for the conjugacy classes of finite subgroups of Γ . Then any simplex $H_0 \subset \cdots \subset H_n$ of $|S|$ is equivalent $\bmod \Gamma$ to one such that $H_n \in \mathcal{F}$. But there are clearly only finitely many such simplices, since H_n is finite. To prove the assertion about the isotropy groups, note that $N(H)$ for any $H \in S$ permutes the simplices $H_0 \subset \cdots \subset H_n$ with $H_n = H$. Since there are only finitely many such simplices, it follows that each such simplex σ has $(N(H) : N(H)_\sigma) < \infty$, hence $N(H)_\sigma$ is of finite homological type. But $N(H)_\sigma = \Gamma_\sigma$, so the first part of the theorem is proved.

Now let X , Γ' , G , and Y be as in the proof of Theorem 9.3. By VIII.11.2 we may assume that X^H is contractible for each $H \in S$. Let X_0 be the part of X where the action of Γ is not free, i.e.

$$X_0 = \bigcup_{H \in S} X^H,$$

and let $Y_0 = X_0/\Gamma' \subseteq Y$. The analysis of the isotropy groups G_i that we gave in the proof of 9.3 shows that Y_0 is the part of Y where the G -action is not free. [Y_0 is in fact equal to the subcomplex called Y_{n-1} in the proof of 10.1.]

I claim now that $H_*(Y_0)$ is finitely generated, and hence so is $H_*(Y, Y_0)$. Accepting this for the moment, we can apply 5.3 to $C(Y, Y_0)$ since G acts freely in $Y - Y_0$. It follows that $\chi(Y, Y_0)$ is divisible by $|G|$. Thus

$$\begin{aligned}\chi(\Gamma) &= \frac{\chi(Y)}{|G|} \\ &= \frac{\chi(Y_0)}{|G|} + \frac{\chi(Y, Y_0)}{|G|} \\ &\equiv \frac{\chi(Y_0)}{|G|} \pmod{\mathbb{Z}}.\end{aligned}$$

To complete the proof, then, we will prove that $H_*(Y_0)$ is finitely generated and that $\chi(Y_0)/|G| = \chi_{\Gamma}(S)$. Note first that $H_*(Y_0) \approx H_{\Gamma}^{\Gamma}(X_0)$ since Γ' acts freely on X_0 (cf. VII.7.8). Next, we apply the following observation to X_0 :

(11.2) Lemma. *Let Z be an admissible proper Γ -complex such that each Γ_{σ} is non-trivial and each Z^H ($H \in S$) is acyclic, where S is the set of non-trivial finite subgroups of Γ . Then Z is homologically equivalent to S , in the following sense: There is a chain complex C of Γ -modules which admits Γ -maps $C(Z) \leftarrow C \rightarrow C(S)$ inducing homology isomorphisms.*

(Note: By “acyclic” we mean “having the homology of a point.” In particular, acyclic $\Rightarrow$ non-empty.)

The proof is a straightforward analogue of that of VII.4.4. Details will be given in the next section. [The intuitive idea is as follows: We have $Z = \bigcup_{H \in S} Z^H$, and the order relation on the index set S completely determines the intersection pattern of the acyclic complexes Z^H . Namely, $Z^{H_1} \cap Z^{H_2} \neq \emptyset$ if and only if H_1 and H_2 have a least upper bound H in S , in which case $Z^{H_1} \cap Z^{H_2} = Z^H$. It is not surprising, then, that we can use $|S|$ instead of the nerve of the covering in VII.4.4.]

Returning to the proof of 11.1, we conclude (cf. VII.5.2) that $H_*(Y_0) \approx H_{\Gamma}^{\Gamma}(X_0) \approx H_{\Gamma}^{\Gamma}(S)$. In view of 7.3c and b', it follows that $H_*(Y_0)$ is finitely generated and that

$$\frac{\chi(Y_0)}{|G|} = \frac{\tilde{\chi}_{\Gamma}(S)}{|G|} = \chi_{\Gamma}(S). \quad \square$$

With very little effort, we can generalize 11.1 to the following result:

(11.3) Proposition. *Let Γ and S be as in 11.1. Let Z be an admissible Γ -complex such that $\chi_{\Gamma}(Z)$ is defined. If Z^H is acyclic for each $H \in S$, then*

$$\chi_{\Gamma}(Z) \equiv \chi_{\Gamma}(S) \pmod{\mathbb{Z}}.$$

(This reduces to 11.1 if Z is a point.)

PROOF. Let X and Γ' be as above, and apply the proof of 11.1 with X replaced by $X \times Z$. Note that the latter is finite dimensional, admissible, and proper, and that $(X \times Z)^H = X^H \times Z^H$ is acyclic for each $H \in S$. Moreover,

$$H_*((X \times Z)/\Gamma') \approx H_*^{\Gamma'}(X \times Z) \approx H_*^{\Gamma'}(Z)$$

since X is contractible. We therefore obtain, as in the proof of 11.1,

$$\begin{aligned} \chi_r(Z) &= \frac{\tilde{\chi}_{\Gamma'}(Z)}{(\Gamma: \Gamma')} \\ &= \frac{\chi((X \times Z)/\Gamma')}{(\Gamma: \Gamma')} \\ &\equiv \frac{\chi((X \times Z)_0/\Gamma')}{(\Gamma: \Gamma')} \pmod{\mathbb{Z}} \\ &= \chi_r(S). \end{aligned} \quad \square$$

Combining 11.1 and 11.3, we see that

$$(11.4) \quad \chi(\Gamma) \equiv \chi_r(Z) \pmod{\mathbb{Z}}$$

under the hypotheses of 11.3.

Finally, we will need the following “local” version of 11.4, for which the hypotheses on Γ can be weakened.

For any prime p , let $\mathbb{Z}_{(p)} = \{a/b: a, b \in \mathbb{Z}, p \nmid b\}$.

(11.5) Proposition. *Let Γ be a group of finite homological type and let Z be an admissible Γ -complex such that $\chi_r(Z)$ is defined. If p is a prime such that Z^H is acyclic for every non-trivial finite p -subgroup of Γ , then*

$$\chi(\Gamma) \equiv \chi_r(Z) \pmod{\mathbb{Z}_{(p)}}.$$

PROOF. We continue to use the notation of the proof of 11.1. Assume first that Γ/Γ' is a p -group. Then every finite subgroup of Γ is a p -group, so our hypothesis says that Z^H is acyclic for every $H \in S$. We now repeat the proofs of 11.1 and 11.3, but taking all homology groups to have $\mathbb{Z}_p$ -coefficients. We no longer have hypotheses to guarantee that $\chi_r(S)$ is defined, but instead we can use VII.10.5 as in the proof of 10.1 to conclude that $H_*(Y_0; \mathbb{Z}_p)$ is finitely generated. Arguing as in the proof of 11.1 (and using 5.7 instead of 5.3), we find that $H_*^{\Gamma'}(S; \mathbb{Z}_p)$ is finitely generated and that

$$\chi(\Gamma) \equiv \frac{\tilde{\chi}_{\Gamma'}(S)}{(\Gamma: \Gamma')} \pmod{\mathbb{Z}},$$

where the equivariant Euler characteristic on the right is now understood to be defined in terms of $H_*^\Gamma(S; \mathbb{Z}_p)$. Similarly,

$$\chi_\Gamma(Z) \equiv \frac{\tilde{\chi}_\Gamma(S)}{(\Gamma: \Gamma')} \pmod{\mathbb{Z}},$$

so

$$\chi(\Gamma) \equiv \chi_\Gamma(Z) \pmod{\mathbb{Z}}.$$

In the general case, where Γ/Γ' is not necessarily a p -group choose a subgroup $\Gamma_p \supseteq \Gamma'$ such that Γ_p/Γ' is a p -Sylow subgroup of Γ/Γ' . By what we have just proved,

$$\chi(\Gamma_p) \equiv \chi_{\Gamma_p}(Z) \pmod{\mathbb{Z}},$$

so that

$$(\Gamma: \Gamma_p) \cdot \chi(\Gamma) \equiv (\Gamma: \Gamma_p) \cdot \chi_{\Gamma_p}(Z) \pmod{\mathbb{Z}}.$$

Since $(\Gamma: \Gamma_p)$ is relatively prime to p , this proves the proposition. $\square$

Knowing $\chi(\Gamma) \pmod{\mathbb{Z}_{(p)}}$ is equivalent to knowing the “ p -fractional part” of $\chi(\Gamma)$, i.e., that part of the partial fractions decomposition of $\chi(\Gamma)$ with denominator a power of p . Thus 11.5 says that $\chi(\Gamma)$ and $\chi_\Gamma(Z)$ have the same p -fractional part. We will give some substance to this result in §13 by giving an interesting example of a complex Z satisfying the hypotheses of 11.5.

12 Acyclic Covers; Proof of Lemma 11.2

The proof of Lemma 11.2 will be based on the following variant of VII.4.4:

(12.1) Proposition. *Let X be a CW-complex and $(X_s)_{s \in S}$ a family of subcomplexes indexed by a partially ordered set S . Assume that $X_s \supseteq X_t$ whenever $s \leq t$. For each cell e of X let $S_e = \{s \in S: e \subseteq X_s\}$. If each subcomplex X_s is acyclic and each partially ordered set S_e is acyclic, then $H_*(X) \approx H_*(S)$. More precisely, there is a chain complex C , constructed canonically from the given data, which admits canonical weak equivalences $C(X) \leftarrow C \rightarrow C(S)$.*

PROOF. We will construct a double complex analogous to that used in §VII.4. Let $|S|^{(p)}$ be the set of p -simplices of $|S|$. For each simplex $\sigma = (s_0 < \cdots < s_p)$ in $|S|^{(p)}$, let $X_\sigma = X_{s_p}$. Then $X_\sigma \subseteq X_\tau$ if τ is a face of σ . In particular, we have inclusion maps

$$\partial_i: C(X_\sigma) \rightarrow C(X_{\partial_i \sigma}) \quad (i = 0, \dots, p),$$

where $\partial_i \sigma = (s_0 < \cdots < \hat{s}_i < \cdots < s_p)$. Let

$$C_p = \bigoplus_{\sigma \in |S|^{(p)}} C(X_\sigma).$$

Letting $\partial = \sum (-1)^i \partial_i: C_p \rightarrow C_{p-1}$, we obtain a chain complex

$$\cdots \rightarrow C_p \xrightarrow{\partial} C_{p-1} \rightarrow \cdots$$

in the category of chain complexes, and hence a double complex. The acyclicity of each X_e implies that one of the spectral sequences of the double complex collapses, yielding

$$H_*(C) \approx H_*(S),$$

where C is the total complex associated to the double complex. On the other hand, exactly as in §VII.4, the acyclicity of each S_e implies that the other spectral sequence also collapses and yields

$$H_*(C) \approx H_*(X).$$

It is easy to check that these isomorphisms are induced by canonical chain maps $C(X) \leftarrow C \rightarrow C(S)$, cf. exercise 1 of §VII.4. $\square$

We will also need the following simple observation:

(12.2) Lemma. *If S is a partially ordered set with a largest or smallest element, then S is contractible.*

PROOF. If $s \in S$ is the extreme element and $S' = S - \{s\}$, then $|S|$ is the cone over $|S'|$, with s as the cone vertex. $\square$

PROOF OF LEMMA 11.2. We have $Z = \bigcup_{H \in S} Z^H$, where S is the set of non-trivial finite subgroups of Γ . For each cell e of Z , the set S_e defined in 12.1 is simply the set of all non-trivial finite subgroups of the isotropy group Γ_e . Since Γ_e is a non-trivial finite group, S_e has a largest element and hence is contractible by 12.2. The hypotheses of 12.1 are therefore satisfied, and we obtain a complex C and weak equivalences $C(Z) \leftarrow C \rightarrow C(S)$. Since Γ acts on Z and permutes the subcomplexes Z^H according to the conjugation action of Γ on S , it is clear that the complex C inherits a Γ -action and that the weak equivalences above are compatible with the Γ -action. $\square$

13 The p -Fractional Part of $\chi(\Gamma)$

Theorem 11.1 is not very useful in practice because the computation of $\chi_\Gamma(S)$ tends to be extremely tedious. If we fix a prime p , however, and replace S by the set of non-trivial finite subgroups which are p -groups, then the computation often simplifies drastically. Moreover, it turns out that this gives the p -fractional part of $\chi(\Gamma)$. In fact, we can even compute the latter by looking

only at the *elementary abelian* p -subgroups of Γ , i.e., the subgroups isomorphic to a finite direct product of copies of $\mathbb{Z}_p$:

(13.1) Theorem. *Let Γ be a group of finite homological type and let $\mathcal{A}$ be the partially ordered set of non-trivial elementary abelian p -subgroups of Γ , where p is a fixed prime. Assume that the normalizer $N(A)$ is of finite homological type for every $A \in \mathcal{A}$. Then $\chi_\Gamma(\mathcal{A})$ is defined and*

$$\chi(\Gamma) \equiv \chi_\Gamma(\mathcal{A}) \pmod{\mathbb{Z}_{(p)}}.$$

(This is an improvement due to Quillen [1978] of the main result of Brown [1975b].)

The proof will use:

(13.2) Lemma. *Let Γ be a group such that $\text{vcd } \Gamma < \infty$ and let p be a prime. If Γ has a torsion-free normal subgroup Γ' of finite index such that $H_*(\Gamma', \mathbb{Z}_p)$ is finitely generated, then Γ has only finitely many p -subgroups, up to conjugacy.*

PROOF. Let X be a finite dimensional, contractible, proper, admissible Γ -complex such that $X^H \neq \emptyset$ for every finite $H \subseteq \Gamma$. Let $Y = X/\Gamma'$ and let $G = \Gamma/\Gamma'$. By hypothesis $H_*(Y; \mathbb{Z}_p)$ is finitely generated, so we know that $H_*(Y^P; \mathbb{Z}_p)$ is finitely generated for every p -subgroup $P \subseteq G$ (§VII.10, Theorem 10.5 and exercise 2). In particular, Y^P has only finitely many connected components.

On the other hand, I claim that

$$Y^P = \coprod_{H \in \mathcal{C}} X^H / (\Gamma' \cap N(H)),$$

where $\mathcal{C}$ is a set of representatives for the Γ' -conjugacy classes of finite subgroups of Γ whose image in G is P . To see this, consider the inverse image $\widetilde{Y}^P$ of Y^P in X , and let $\widetilde{\mathcal{C}}$ be the set of finite subgroups of Γ whose image in G is P . For any $x \in \widetilde{Y}^P$, we know from §10 that the isotropy group Γ_x maps isomorphically to $G_y \subseteq G$, where $y \in Y$ is the image of x . Since $P \subseteq G_y$, it follows that there is a unique $H \subseteq \Gamma_x$ whose image in G is P . In other words, there is a unique $H \in \widetilde{\mathcal{C}}$ such that $x \in X^H$. Thus $\widetilde{Y}^P = \coprod_{H \in \widetilde{\mathcal{C}}} X^H$. Since Γ' permutes the X^H according to its conjugation action on $\widetilde{\mathcal{C}}$, it follows that

$$Y^P = \left(\coprod_{H \in \widetilde{\mathcal{C}}} X^H \right) / \Gamma' = \coprod_{H \in \mathcal{C}} X^H / (\Gamma' \cap N(H)),$$

as claimed.

Since Y^P has only finitely many components and each X^H is non-empty, we conclude that $\mathcal{C}$ is finite, i.e., that the finite subgroups of Γ lying over P fall into finitely many conjugacy classes. Finally, there are only finitely many possibilities for P , so the lemma follows. $\square$

We also need:

(13.3) Lemma. *Let S and T be partially ordered sets and let $f_0, f_1: S \rightarrow T$ be order-preserving maps such that $f_0(s) \leq f_1(s)$ for all $s \in S$. Then $|f_0| \simeq |f_1|: |S| \rightarrow |T|$.*

[Note that this reduces to 12.2 if $S = T$ and either f_0 or f_1 is id_S while the other is a constant map.]

PROOF. Heuristically, the desired homotopy moves $f_0(s)$ to $f_1(s)$ along the 1-simplex $\{f_0(s), f_1(s)\}$ of $|T|$. Unfortunately, this only defines the homotopy on vertices of $|S|$, so we instead proceed as follows. Let J be the ordered set $\{0, 1\}$ with $0 < 1$. Note that $|J|$ is the unit interval. Consider $S \times J$ with the product ordering: $(s, i) \leq (s', i')$ if and only if $s \leq s'$ and $i \leq i'$. Let $F: S \times J \rightarrow T$ be defined by $F(s, i) = f_i(s)$. The hypothesis that $f_0(s) \leq f_1(s)$ guarantees that F is order-preserving, so we have an induced map $|F|: |S \times J| \rightarrow |T|$. Now it is immediate from the definitions that $|S_1 \times S_2| = |S_1| \times |S_2|$ for any two partially ordered sets S_1, S_2 , where the product on the right is the simplicial product defined in §VIII.11. Thus $|F|$ can be viewed as a map $|S| \times |J| \rightarrow |T|$, and this is the required homotopy. $\square$

PROOF OF THEOREM 13.1. By Lemma 13.2, Γ has only finitely many p -subgroups, up to conjugacy. It now follows exactly as in the first part of the proof of 11.1 that $\chi_r(\mathcal{A})$ is defined. We now wish to apply 11.5 to the Γ -complex $|\mathcal{A}|$, so we must verify that $|\mathcal{A}|^H$ is acyclic for each non-trivial p -subgroup $H \subseteq \Gamma$. Now $|\mathcal{A}|^H = |\mathcal{A}^H|$, where $\mathcal{A}^H$ is the set of non-trivial elementary abelian p -groups normalized by H , and I claim that this is in fact contractible. For let $C \subseteq H$ be a central subgroup of order p , and consider for any $A \in \mathcal{A}^H$ the chain of inequalities

$$A \geq A^C \leq C \cdot A^C \leq C$$

in $\mathcal{A}^H$. (Note that A^C , C , and $C \cdot A^C$ are indeed all in $\mathcal{A}^H$; in the case of A^C , this follows from VI.8.1.) This yields, via 13.3, a homotopy from the identity map of $\mathcal{A}^H$ to the constant map $A \mapsto C$, since the intermediate maps $A \mapsto A^C$ and $A \mapsto C \cdot A^C$ are order-preserving maps $\mathcal{A} \rightarrow \mathcal{A}$. Thus 11.5 is applicable and yields the desired congruence. $\square$

In the next section we will show that $\chi_r(\mathcal{A})$ admits an explicit computation in terms of the normalizers of the elements $A \in \mathcal{A}$. For the moment, we just mention one case where this is obvious. Suppose that every $A \in \mathcal{A}$ is of rank ≤ 1 . [This is equivalent to saying that the finite subgroups of Γ have p -periodic cohomology, cf. VI.9.7. It is also equivalent, as we will see in the next chapter, to saying that Γ itself has p -periodic cohomology in high dimensions.] In this case $|\mathcal{A}|$ is discrete and we obviously have

$$\chi_r(\mathcal{A}) = \sum_{P \in \mathcal{P}} \chi(N(P)),$$

where $\mathcal{P}$ is a set of representatives for the conjugacy classes of subgroups of Γ of order p . Thus

$$(13.4) \quad \chi(\Gamma) \equiv \sum_{P \in \mathcal{P}} \chi(N(P)) \pmod{\mathbb{Z}_{(p)}}$$

in this case.

For computational purposes, it is often more convenient to work with elements of order p and their centralizers, rather than subgroups of order p and their normalizers. It is easy to rewrite 13.4 in these terms:

(13.5) Corollary. *Let Γ be a group of finite homological type and assume that every elementary abelian p -subgroup of Γ has rank ≤ 1 . Suppose further that the centralizer $C(\gamma)$ is of finite homological type for every $\gamma \in \Gamma$ of order p , and let $\mathcal{C}$ be a set of representatives for the conjugacy classes of elements of Γ of order p . Then $\mathcal{C}$ is finite and*

$$\chi(\Gamma) \equiv \frac{1}{p-1} \sum_{\gamma \in \mathcal{C}} \chi(C(\gamma)) \pmod{\mathbb{Z}_{(p)}}.$$

PROOF. For each $P \in \mathcal{P}$ choose a set $\mathcal{C}_P$ of representatives for the non-trivial elements of P under the conjugation action of $N(P)/C(P)$. Then $\coprod_{P \in \mathcal{P}} \mathcal{C}_P$ is a set of representatives for the conjugacy classes of elements of Γ of order p , so we may assume that $\mathcal{C}$ is equal to this union. Since $C(P) = C(\gamma)$ for every nontrivial $\gamma \in P$, the action of $N(P)/C(P)$ on $P - \{1\}$ is free. Thus

$$\text{card}(\mathcal{C}_P) \cdot (N(P) : C(P)) = p - 1.$$

Consequently,

$$\begin{aligned} \sum_{\gamma \in \mathcal{C}} \chi(C(\gamma)) &= \sum_{P \in \mathcal{P}} \text{card}(\mathcal{C}_P) \cdot \chi(C(P)) \\ &= \sum_{P \in \mathcal{P}} \text{card}(\mathcal{C}_P) \cdot (N(P) : C(P)) \cdot \chi(N(P)) \\ &= (p-1) \sum_{P \in \mathcal{P}} \chi(N(P)). \end{aligned}$$

The corollary now follows from 13.4. □

We mention briefly one interesting application of 13.5 to number theory. (This application is due to Serre, and a detailed treatment is given in Brown [1974, 1975b].) Fix an odd prime p and let Γ be the symplectic group $Sp_{2n}(\mathbb{Z})$, where $2n = p - 1$.

On the one hand, we have Harder's formula (8.6) which gives $\chi(\Gamma)$ in terms of $\zeta(-1), \dots, \zeta(1 - 2n)$, and hence in terms of the Bernoulli numbers $B_2, \dots, B_{p-1}$. Using known facts about Bernoulli numbers ("von Staudt's Theorem") one concludes that $\chi(\Gamma) \equiv 0 \pmod{\mathbb{Z}_{(p)}}$ if and only if p divides the numerator of one of the numbers $B_2, \dots, B_{p-3}$.

On the other hand, it is not hard to classify the elements of Γ of order p up to conjugacy and to compute their centralizers. One finds (i) that the hypotheses of 13.5 are satisfied; (ii) that the number of conjugacy classes of elements of order p is closely related to the class number h of the p -th cyclotomic number field; and (iii) that the centralizer of any element γ of order p is finite and of order $2p$, so that $\chi(C(\gamma)) = 1/2p$. It then follows from 13.5 that $\chi(\Gamma) \equiv 0 \pmod{\mathbb{Z}_{(p)}}$ if and only if $h/2p \equiv 0 \pmod{\mathbb{Z}_{(p)}}$, i.e., if and only if $p|h$. (The prime p is then said to be *irregular*.)

Combining the results of the last two paragraphs, we have: A prime p is irregular if and only if p divides the numerator of one of the Bernoulli numbers $B_2, \dots, B_{p-3}$. This result is not new; indeed, it is known as *Kummer's criterion* for regularity. But our method of proof via Euler characteristics generalizes to rings of integers other than $\mathbb{Z}$ and yields number-theoretic results which were not previously known.

14 A Formula for $\chi_\Gamma(\mathcal{A})$

The formula to be given in this section is essentially due to Quillen [private communication]. It is based on:

(14.1) Proposition. *Let V be a vector space of dimension $r \geq 1$ over the field $\mathbb{F}_q$ with q elements, and let $S = S(V)$ be the partially ordered set of non-trivial proper subspaces of V . Then*

$$\tilde{H}_i(S) \approx \begin{cases} \mathbb{Z}^{n(r)} & \text{if } i = r - 2 \\ 0 & \text{if } i \neq r - 2, \end{cases}$$

where $n(r) = q^{r(r-1)/2}$.

(This is a special case of the so-called "Solomon-Tits theorem." The proof that we will give is due to Quillen [1973]. It will follow easily from the proof that $|S|$ in fact has the homotopy type of a bouquet of $n(r)$ spheres of dimension $r - 2$, but we will not need to know this.)

PROOF. The proposition is trivially true if $r = 1$, since S is then empty. Assume now that $r > 1$ and that the proposition is known for vector spaces of dimension $r - 1$. Let $L \subset V$ be a fixed 1-dimensional subspace and let $S' \subset S$ be the set of non-trivial subspaces $V' \subset V$ such that $V' + L \neq V$ (i.e., such that $V' + L \in S$). For any $V' \in S'$, we have relations

$$V' \leq V' + L \leq L$$

in S' , and it follows easily that S' is contractible (cf. proof of 13.1). Thus

$$\tilde{H}_i(S) \approx H_i(S, S').$$

Let $\mathcal{H}$ be the set of hyperplanes H in V which are complementary to L (i.e., such that $V = L \oplus H$). Let $\bar{S}(H)$ (resp. $S(H)$) be the set of non-trivial subspaces (resp. non-trivial proper subspaces) of H . Note that every simplex σ of $|S|$ which is not in $|S'|$ has the form

$$V_0 < \cdots < V_n,$$

where $V_n \in \mathcal{H}$. Hence σ is a simplex of $|\bar{S}(H)|$ but not $|S(H)|$ for a unique $H \in \mathcal{H}$. Consequently, there is an excision isomorphism (already on the level of simplicial chain complexes)

$$H_*(S, S') \xrightarrow{\approx} \bigoplus_{H \in \mathcal{H}} H_*(\bar{S}(H), S(H)).$$

Now $\bar{S}(H)$ is contractible since it has a largest element, so

$$H_i(\bar{S}(H), S(H)) \approx \tilde{H}_{i-1}(S(H)) \approx \begin{cases} \mathbb{Z}^{n(r-1)} & \text{if } i = r - 2 \\ 0 & \text{if } i \neq r - 2 \end{cases}$$

by the induction hypothesis. Thus

$$\tilde{H}_i(S) \approx \begin{cases} \mathbb{Z}^{\text{card}(\mathcal{H}) \cdot n(r-1)} & \text{if } i = r - 2 \\ 0 & \text{if } i \neq r - 2. \end{cases}$$

To compute $\text{card}(\mathcal{H})$, note that the elements of $\mathcal{H}$ correspond to splittings of the exact sequence

$$0 \rightarrow L \rightarrow V \rightarrow V/L \rightarrow 0,$$

so there is a bijection $\mathcal{H} \approx \text{Hom}_{\mathbb{F}_q}(V/L, L)$. The latter being a vector space of dimension $r - 1$, it follows that $\text{card}(\mathcal{H}) = q^{r-1}$. One now completes the proof by verifying that the function $n(r) = q^{r(r-1)/2}$ satisfies $n(r) = q^{r-1}n(r-1)$. $\square$

(14.2) Theorem. Let Γ be a group and p a prime such that $\chi_r(\mathcal{A})$ is defined, where $\mathcal{A}$ is the set of non-trivial elementary abelian p -subgroups of Γ . Let $\mathcal{A}_0$ be a set of representatives for $\mathcal{A} \bmod \Gamma$. Then

$$\chi_r(\mathcal{A}) = \sum_{A \in \mathcal{A}_0} (-1)^{r(A)-1} p^{n(A)} \chi(N(A)),$$

where $r(A)$ is the rank of A and $n(A) = r(A)(r(A) - 1)/2$. Consequently,

$$\chi(\Gamma) \equiv \sum_{A \in \mathcal{A}_0} (-1)^{r(A)-1} p^{n(A)} \chi(N(A)) \pmod{\mathbb{Z}_{(p)}}$$

under the hypotheses of 13.1.

PROOF. For each $A \in \mathcal{A}$, let $\bar{S}(A)$ (resp. $S(A)$) be the set of non-trivial subgroups (resp. non-trivial proper subgroups) of A . Let $\mathcal{S}(A)$ be a set of representatives $\bmod N(A)$ for the simplices of $|\bar{S}(A)|$ which are not in $|S(A)|$. Then

$\coprod_{A \in \mathcal{A}_0} \mathcal{S}(A)$ is a set of representatives for the simplices of $|\mathcal{A}| \bmod \Gamma$. It therefore follows from the definition of $\chi_\Gamma(\mathcal{A})$ that

$$\chi_\Gamma(\mathcal{A}) = \sum_{A \in \mathcal{A}_0} \chi_{N(A)}(\bar{S}(A), S(A)),$$

where the relative equivariant Euler characteristic on the right is equal, by definition, to $\sum_{\sigma \in \mathcal{S}(A)} (-1)^{\dim \sigma} \chi(N(A)_\sigma)$.

I claim that $\chi_{N(A)}(\bar{S}(A), S(A)) = \chi(N(A)) \cdot \chi(\bar{S}(A), S(A))$. For let $C \subseteq N(A)$ be a torsion-free subgroup of finite index which centralizes A . Then C acts trivially on $\bar{S}(A)$, so we have a spectral sequence (cf. VII.7.2)

$$E_{pq}^2 = H_p(C, \mathbb{Q}) \otimes_{\mathbb{Q}} H_q(\bar{S}(A), S(A); \mathbb{Q}) \Rightarrow H_{p+q}^C(\bar{S}(A), S(A); \mathbb{Q}).$$

Therefore $\tilde{\chi}_C(\bar{S}(A), S(A)) = \tilde{\chi}(C) \cdot \chi(\bar{S}(A), S(A))$. The claim now follows at once from 7.3a and b and from relative versions of 7.3b' and c.

Thus

$$\chi_\Gamma(\mathcal{A}) = \sum_{A \in \mathcal{A}_0} \chi(N(A)) \cdot \chi(\bar{S}(A), S(A)).$$

Now A can be regarded as a vector space of dimension $r = r(A)$ over $\mathbb{F}_p$, so we can apply 14.1. Since $\bar{S}(A)$ is contractible, we conclude that $\chi(\bar{S}(A), S(A)) = (-1)^{r-1} p^{r(r-1)/2}$, whence the theorem. $\square$

Remark. Let p^a be the maximal order of a p -subgroup of Γ . Then the denominators of the $\chi(N(A))$ which occur in the theorem involve p to at most the a -th power by 9.3. Since $r(r-1)/2 \geq 1$ if $r \geq 2$, it follows from 14.2 that

$$\chi(\Gamma) \equiv \sum_{P \in \mathcal{P}} \chi(N(P)) \pmod{p^{-(a-1)} \mathbb{Z}_{(p)}},$$

where $\mathcal{P}$ is a set of representatives for the subgroups of Γ of order p . Thus we can compute the "leading term" of the partial fractions decomposition of $\chi(\Gamma)$ (i.e., the term involving p^a in the denominator) by considering only the subgroups of order p and their normalizers.

EXERCISES

1. Extract from the proof of 14.2 the following fact: Let S be a partially ordered Γ -set such that $\chi_\Gamma(S)$ is defined. For each $s \in S$, let $S_{\leq s}$ (resp. $S_{< s}$) be the set of $t \in S$ such that $t \leq s$ (resp. $t < s$). If each $S_{\leq s}$ is finite, then

$$\chi_\Gamma(S) = \sum_s \chi(\Gamma_s) \cdot \chi(S_{\leq s}, S_{< s}),$$

where s ranges over a set of representatives for $S \bmod \Gamma$.

2. Show that the hypothesis that each $S_{< s}$ is finite in exercise 1 can be replaced by the weaker hypothesis that $H_\bullet(S_{< s})$ is finitely generated for each s . [Hint: Γ_s has a subgroup of finite index which acts trivially on $H_\bullet(S_{\leq s}, S_{< s}; \mathbb{Z}_2)$.]

CHAPTER X

Farrell Cohomology Theory

1 Introduction

Let Γ be a group such that $\text{vcd } \Gamma < \infty$. If Γ is torsion-free, then we know from Chapter VIII that $\text{cd } \Gamma < \infty$, so that Γ has no high-dimensional cohomology. In the general case, one might hope to “explain” the high-dimensional cohomology of Γ in terms of the torsion in Γ . (This is analogous to the situation of Chapter IX, where we tried to explain the non-integrality of $\chi(\Gamma)$ in terms of the torsion in Γ .)

For this purpose it is convenient to use modified cohomology groups $\hat{H}^*(\Gamma, M)$, first introduced by Farrell [1977]. Farrell’s theory generalizes the Tate cohomology theory for finite groups (Chapter VI), and it has the following two properties which make it appropriate for our present purposes: (i) $\hat{H}^i = H^i$ for $i > \text{vcd } \Gamma$ and (ii) $\hat{H}^* = 0$ if Γ is torsion-free.

The next three sections will be devoted to the foundations of the Farrell cohomology theory. Then in §§5 and 6 we will generalize to the Farrell theory some of the results of §§VI.8 and VI.9 on cohomological triviality and periodicity. Finally, §7 will contain the main results of the chapter, relating $\hat{H}^*(\Gamma)$ to the torsion in Γ .

2 Complete Resolutions

Let $\text{vcd } \Gamma = n < \infty$. By a *complete resolution* for Γ we mean an acyclic chain complex F of projective $\mathbb{Z}\Gamma$ -modules, together with an ordinary projective resolution $\varepsilon: P \rightarrow \mathbb{Z}$ of $\mathbb{Z}$ over $\mathbb{Z}\Gamma$ such that F and P coincide in sufficiently high dimensions. (As in Chapter VI, F is *not* in general

nonnegative.) If Γ is finite, for example, then any complete resolution in the sense of §VI.3 is also a complete resolution in the present sense. In §VI.3, however, we required F and P to agree in *all* non-negative dimensions, so the present notion of complete resolution is more general. (It might seem reasonable, by analogy with §VI.3, to require F and P to agree in dimensions $\geq n = \text{vcd } \Gamma$. Indeed, we will see below that complete resolutions of this type always exist. Our reason for not requiring this is that we want a complete resolution for Γ to also be a complete resolution for any subgroup of Γ .)

If (F, P, ε) and (F', P', ε') are complete resolutions, then a chain map $\tau: F \rightarrow F'$ will be called a *map of complete resolutions* if there is an augmentation-preserving map $P \rightarrow P'$ which agrees with τ in sufficiently high dimensions.

(2.1) Proposition (a) *There exist complete resolutions (F, P, ε) such that F coincides with P in dimensions $\geq n = \text{vcd } \Gamma$. If Γ is of type VFP , then F can be taken to be of finite type.*

(b) *If (F, P, ε) and (F', P', ε') are complete resolutions, then there is a unique homotopy class of maps from (F, P, ε) to (F', P', ε') , and these maps are homotopy equivalences.*

PROOF. Choose a torsion-free subgroup $\Gamma' \subseteq \Gamma$ of finite index. We will apply the relative homological algebra of §VI.2 to (Γ, Γ') . Thus we have a notion of *admissible* short exact sequence of Γ -modules and a corresponding notion of *relative injective* Γ -module. Let $\varepsilon: P \rightarrow \mathbb{Z}$ be a projective resolution of $\mathbb{Z}$ over $\mathbb{Z}\Gamma$ and let $K = \ker\{P_{n-1} \rightarrow P_{n-2}\}$. Then $(P_i)_{i \geq n}$ provides a projective resolution

$$\cdots \rightarrow P_{n+1} \rightarrow P_n \rightarrow K \rightarrow 0$$

of K . Since $\text{cd } \Gamma' = n$, we know (VIII.2.1) that K is $\mathbb{Z}\Gamma'$ -projective. Proposition VI.2.6 therefore gives us a relative injective resolution

$$0 \rightarrow K \rightarrow Q^0 \rightarrow Q^1 \rightarrow \cdots$$

with each Q^i projective. Splicing together $(P_i)_{i \geq n}$ and $(Q^i)_{i \geq 0}$, we obtain the desired complete resolution. Note that P and Q can be taken to be of finite type if Γ is of type VFP , whence (a).

The proof of (b) will require the following generalization of VI.3.2:

(2.2) Lemma. *If $\text{cd } \Gamma < \infty$ then any acyclic chain complex of projective $\mathbb{Z}\Gamma$ -modules is contractible.*

PROOF. Note first that any Γ -module M which is $\mathbb{Z}$ -free has $\text{proj dim}_{\mathbb{Z}\Gamma} M \leq n = \text{cd } \Gamma$. For if P is a projective resolution of $\mathbb{Z}$ of length n , then $P \otimes M$ (with diagonal Γ -action) is a projective resolution of $\mathbb{Z} \otimes M = M$ of length n . [Why?] Suppose now that F is an acyclic complex of projectives and let Z_k for any k be the module of k -cycles. Then we have an exact sequence

$$0 \rightarrow Z_k \rightarrow F_k \rightarrow \cdots \rightarrow F_{k-n+1} \rightarrow Z_{k-n} \rightarrow 0$$

with each F_i projective. Since $\text{proj dim}_{\mathbf{Z}\Gamma} Z_{k-n} \leq n$, it follows (VIII.2.1) that Z_k is projective. Hence $F_{k+1} \rightarrow Z_k$ splits for all k and F is contractible. $\square$

Returning now to the proof of 2.1b, the lemma shows that any complete resolution F is Γ' -contractible, hence it is admissible. Also, each projective module F_i is relatively injective by VI.2.3. Given two complete resolutions as in (b), we use ordinary homological algebra to construct an augmentation-preserving map $P \rightarrow P'$; this defines a chain map $\tau: F \rightarrow F'$ in high dimensions. Since F is acyclic and admissible and F' is relatively injective, relative homological algebra (VI.2.4) now allows us to extend τ to all dimensions. Similarly, any two such maps are homotopic in high dimensions by ordinary homological algebra applied to P and P' , and the homotopy can be extended to all dimensions by relative homological algebra. It is clear from the uniqueness that all maps of complete resolutions are homotopy equivalences. $\square$

Various other chain maps can be constructed by the same technique. For instance:

(2.3) Proposition. *If (F, P, ε) is a complete resolution, then there is a unique homotopy class of chain maps $\tau: F \rightarrow P$ such that τ is the identity in sufficiently high dimensions.*

PROOF. F is acyclic and admissible and each P_i is relatively injective. $\square$

In addition, one can construct a “diagonal map,” which will be needed for the theory of cup products:

(2.4) Proposition. *Let (F, P, ε) be a complete resolution and let $\tau: P \rightarrow P \otimes P$ be a diagonal approximation (§V.1) with components $\tau_{pq}: P_{p+q} \rightarrow P_p \otimes P_q$. Let $F \hat{\otimes} F$ be the completed tensor product as in §VI.5. If m is an integer such that F and P agree in dimensions $\geq m-1$, then there is a chain map $\Delta: F \rightarrow F \hat{\otimes} F$ whose (m, m) -component $\Delta_{mm}: F_{2m} \rightarrow F_m \otimes F_m$ is equal to τ_{mm} .*

PROOF. This is almost identical to the proof of the corresponding result for finite groups (end of §VI.5). As in that case, it suffices to define Δ in dimension $2m$, in such a way that $\partial\Delta|_{B_{2m}} = 0$ and $\Delta_{mm} = \tau_{mm}$. This reduces to the construction of a family of maps $\alpha_p: F_{2m} \rightarrow F_{m+p} \otimes F_{m-p}$ ($p \in \mathbb{Z}$) satisfying (i) $(\partial'\alpha_p + \partial''\alpha_{p-1})|_{B_{2m}} = 0$ and (ii) $\alpha_0 = \tau_{mm}$. (Here ∂' , ∂'' , and B_{2m} are as in §VI.5.) Let $\alpha_0 = \tau_{mm}$ and $\alpha_1 = \tau_{m+1, m-1}$; this makes sense because $F = P$ in dimensions $\geq m-1$. Since $\tau: P \rightarrow P \otimes P$ is a chain map, condition (i) (with $p = 1$) is satisfied. We can therefore define α_p inductively for all p , exactly as in §VI.5. $\square$

Finally, recall that in the finite group case it is possible to construct a complete resolution by splicing together an ordinary resolution and its dual.

It turns out that there is an analogous (but more complicated) construction whenever Γ is of type VFP . For simplicity we will treat here only the case where Γ is a virtual duality group, referring to the exercises below for the general case.

Let Γ be a virtual duality group with “dualizing module” $D = H^n(\Gamma, \mathbb{Z}\Gamma)$ (cf. VIII.11.3). I claim that D is of type FP_∞ as a $\mathbb{Z}\Gamma$ -module. For let $\Gamma' \subseteq \Gamma$ be a torsion-free subgroup of finite index and recall that D is of type FP as a $\mathbb{Z}\Gamma'$ -module (§VIII.10, exercise 1); it now follows as in the proof of VIII.5.1 that D is of type FP_∞ over $\mathbb{Z}\Gamma$.

(2.5) Proposition. *Let Γ be a virtual duality group with dualizing module D , let $\varepsilon: P \rightarrow \mathbb{Z}$ be a finite type projective resolution of $\mathbb{Z}$ over $\mathbb{Z}\Gamma$, and let $\eta: Q \rightarrow D$ be a finite type projective resolution of D over $\mathbb{Z}\Gamma$. Then there is a complete resolution F such that the dual complex $\bar{F} = \mathcal{H}om_{\mathbb{Z}\Gamma}(F, \mathbb{Z}\Gamma)$ is the mapping cone of a chain map $\Sigma^{-n}Q \rightarrow \bar{P}$. In particular, $F_i = P_i$ for $i \geq n$ and $F_i = (Q_{n-1-i})^*$ for $i \leq -1$.*

PROOF Consider the n -th suspension $\Sigma^n \bar{P}$ of the dual $\bar{P}$ of P . Then $H_i(\Sigma^n \bar{P}) = H_{i-n}(\bar{P}) = H^{n-i}(\Gamma, \mathbb{Z}\Gamma)$, which is 0 for $i \neq 0$ and D for $i = 0$. It follows easily that there is a weak equivalence $Q \rightarrow \Sigma^n \bar{P}$. [Start by lifting $\eta: Q_0 \rightarrow D$ to a map to the zero-cycles of $\Sigma^n \bar{P}$; now extend this to a chain map by the fundamental lemma I.7.4.] This map can also be viewed as a weak equivalence $f: \Sigma^{-n}Q \rightarrow \bar{P}$, and the mapping cone C of f is an acyclic complex of finitely generated projectives. Let F be the dual complex $\bar{C}$. Thus $F_i = (C_{-i})^* = (\bar{P}_{-i} \oplus (\Sigma^{-n}Q)_{-i-1})^* = P_i^{**} \oplus (Q_{n-1-i})^* = P_i \oplus (Q_{n-1-i})^*$; in particular, $F_i = P_i$ for $i \geq n$. Checking the definition of the differential in F , one finds that it agrees with that in P in dimensions $\geq n$, up to sign. The proposition will be proved, then, if we verify that F is acyclic. Let $\Gamma' \subseteq \Gamma$ be a torsion-free subgroup of finite index. Then we have $F = \mathcal{H}om_{\mathbb{Z}\Gamma}(C, \mathbb{Z}\Gamma) \approx \mathcal{H}om_{\mathbb{Z}\Gamma'}(C, \mathbb{Z}\Gamma')$. [This follows from the theory of induced and coinduced modules, exactly as in the proof of VI.3.4; alternatively, use Lemma VIII.7.4.] But C is Γ' -contractible (by 2.2 for instance), so its Γ' -dual $\mathcal{H}om_{\mathbb{Z}\Gamma'}(C, \mathbb{Z}\Gamma')$ is also Γ' -contractible and hence acyclic. $\square$

Remark. Consider the complex $C = \bar{F}$ of this proof. It coincides with $\Sigma^{1-n}Q$ in dimensions ≥ 1 , so $\Sigma^{n-1}C$ coincides with Q in dimensions $\geq n$. Thus $\Sigma^{n-1}C = \Sigma^{n-1}\bar{F}$ is what one would reasonably call a complete resolution of D .

EXERCISES

1. Let $\Gamma = \Gamma_1 \times \Gamma_2$, where $\text{cd } \Gamma_1 < \infty$ and $\text{vcd } \Gamma_2 < \infty$. Show that one can construct a complete resolution for Γ by taking the tensor product of a finite length projective resolution of $\mathbb{Z}$ over $\mathbb{Z}\Gamma_1$ and a complete resolution for Γ_2 .

*2. A non-negative chain complex C is said to be of *type* FP_∞ if it admits a weak equivalence $P \rightarrow C$, where P is a non-negative chain complex of finitely generated projectives. (If C consists of a single module M concentrated in dimension 0, for example, this just means that M is of type FP_∞ .) If Γ is a group and Γ' is a subgroup of finite index, prove that a chain complex of $\mathbb{Z}\Gamma$ -modules is of type FP_∞ over $\mathbb{Z}\Gamma$ if and only if it is of type FP_∞ over $\mathbb{Z}\Gamma'$. [This is proved in Brown [1982]. All the essential ideas can be found in Brown [1975a], and one can even deduce the result from Theorem 2 of the latter by a “Shapiro’s lemma” argument.]

*3. Suppose Γ is of type VFP and let $\bar{P}$ be as in 2.5. Let $\mathcal{D}$ be the subcomplex

$$0 \rightarrow \bar{P}^0 \rightarrow \dots \rightarrow \bar{P}^{n-1} \rightarrow \mathbb{Z}^n \rightarrow 0$$

of $\bar{P}$, where $\mathbb{Z}^n$ is the module of n -cocycles. Note that the inclusion $\mathcal{D} \hookrightarrow \bar{P}$ is a weak equivalence. Show that $\mathcal{D}$ is of type FP_∞ and deduce that 2.5 extends to groups of type VFP , with Q now taken to be a finite type “projective resolution” of $\Sigma^n \mathcal{D}$, i.e., a finite type complex of projectives such that there is a weak equivalence $Q \rightarrow \Sigma^n \mathcal{D}$.

3 Definition and Properties of $\hat{H}^*(\Gamma)$

We continue to assume that Γ is a group such that $\text{vcd } \Gamma = n < \infty$. We can then choose a complete resolution (F, P, ε) and set

$$\hat{H}^*(\Gamma, M) = H^*(\mathcal{H}om_\Gamma(F, M))$$

for any Γ -module M . By 2.1, $\hat{H}^*$ is well-defined up to canonical isomorphism. If Γ is finite, the present definition is obviously consistent with that of Chapter VI. For simplicity we will often write $\hat{H}^*(\Gamma)$ or even $\hat{H}^*$ instead of $\hat{H}^*(\Gamma, M)$. It will always be understood, however, that there is an arbitrary Γ -module of coefficients.

The cohomology theory $\hat{H}^*(\Gamma, -)$ will be called the *Farrell cohomology theory*. It has formal properties analogous to those of the Tate theory for finite groups (in which case $n = 0$):

(3.1) $\hat{H}^*(\Gamma) = 0$ if Γ is torsion-free.

For in this case we can take $F = 0$.

(3.2) $\hat{H}^*(\Gamma, -)$ has all the “usual” cohomological properties: long exact sequences, Shapiro’s lemma, restriction and transfer maps, and cup products.

This is proved exactly as in ordinary cohomology theory, except for the construction of cup products. We will discuss the latter below, after stating two more properties.

(3.3) $\hat{H}^*(\Gamma, M) = 0$ if M is an induced module $\mathbb{Z}\Gamma \otimes_{\mathbb{Z}\Gamma'} M'$, where Γ' is a torsion-free subgroup of finite index and M' is an arbitrary Γ' -module. Consequently, the functors $\hat{H}^i(\Gamma, -)$ are effaceable and coeffaceable.

This follows from 3.1 and 3.2 (Shapiro's lemma). Note that 3.2 and 3.3 allow us to shift dimensions in both directions, as in VI.5.4. In view of the next property, this means that questions about Farrell cohomology can often be reduced to questions about (high-dimensional) ordinary cohomology.

(3.4) There is a canonical map $H^i \rightarrow \hat{H}^i$ which is an isomorphism for $i > n$ and an epimorphism for $i = n$. Moreover, the sequence

$$H^n(\Gamma') \xrightarrow{\text{tr}} H^n(\Gamma) \longrightarrow \hat{H}^n(\Gamma) \longrightarrow 0$$

is exact for any torsion-free subgroup $\Gamma' \subseteq \Gamma$ of finite index, where tr is the transfer map.

The map $H^i \rightarrow \hat{H}^i$ is induced by the chain map $F \rightarrow P$ of 2.3. Since the latter can be taken to be the identity in dimensions $\geq n$ (cf. 2.1), the first assertion of 3.4 is clear. To prove the second assertion, one could simply directly examine the n -coboundaries in $\mathcal{H}om_\Gamma(F, M)$, where F is assumed to be constructed from P as in the proof of 2.1. It is easier, however, to proceed as follows: Consider the "dimension-shifting" exact sequence

$$0 \rightarrow K \rightarrow \mathbb{Z}\Gamma \otimes_{\mathbb{Z}\Gamma'} M \rightarrow M \rightarrow 0.$$

This gives rise to a long exact sequence in ordinary cohomology which maps to the corresponding sequence in Farrell cohomology. Using Shapiro's lemma and definition (A) of the transfer map (§III.9), we deduce a diagram

$$\begin{array}{ccccccc} H^n(\Gamma', M) & \xrightarrow{\text{tr}} & H^n(\Gamma, M) & \longrightarrow & H^{n+1}(\Gamma, K) & \longrightarrow & 0 \\ & & \downarrow & & \downarrow \approx & & \\ 0 & \longrightarrow & \hat{H}^n(\Gamma, M) & \xrightarrow{\approx} & \hat{H}^{n+1}(\Gamma, K) & \longrightarrow & 0 \end{array}$$

with exact rows. (The zero in the top row is $H^{n+1}(\Gamma', M)$, which vanishes because $\text{cd } \Gamma' = n$.) The second assertion of 3.4 follows at once.

Remarks

1. If Γ is finite, 3.4 reduces to the statement that $H^i = \hat{H}^i$ for $i > 0$ and that H^0 is the cokernel of the norm map $M \rightarrow M^G$.

2. It follows from 3.4 that the image of $\text{tr}: H^n(\Gamma') \rightarrow H^n(\Gamma)$ is independent of the choice of Γ' . This should not surprise the reader who has done exercise 6 of §VIII.2.

We can now explain the construction of cup products. Let (F, P, ε) be a complete resolution and choose $\Delta: F \rightarrow F \otimes F$ as in 2.4. Using Δ we can construct cup products

$$\hat{H}^p(\Gamma, M) \otimes \hat{H}^q(\Gamma, N) \rightarrow \hat{H}^{p+q}(\Gamma, M \otimes N)$$

in the usual way, and these will be compatible with coboundary maps in long exact sequences. Moreover, we have by 2.4 an integer m , such that

$H^i \cong \hat{H}^i$ for $i \geq m$ and such that the cup product $\hat{H}^m \otimes \hat{H}^m \rightarrow \hat{H}^{2m}$ agrees with the ordinary cup product $H^m \otimes H^m \rightarrow H^{2m}$. The standard dimension-shifting arguments (cf. proof of VI.5.8) now show that the cup product in $\hat{H}^*$ is compatible with that in H^* , in the sense that the diagram

$$\begin{array}{ccc} H^p \otimes H^q & \xrightarrow{\cup} & H^{p+q} \\ \downarrow & & \downarrow \\ \hat{H}^p \otimes \hat{H}^q & \xrightarrow{\cup} & \hat{H}^{p+q} \end{array}$$

commutes for all p, q . [Assuming this for (p, q) , use the coeffaceability of H^* to deduce it for $(p+1, q)$ and $(p, q+1)$, and use the effaceability of $\hat{H}^*$ to deduce it for $(p-1, q)$ and $(p, q-1)$.]

It is now clear that our cup product is independent of the choice of F and Δ . For any two choices would give the same product in high dimensions by what we have just done, and hence they would agree in all dimensions by dimension-shifting. Finally, one uses dimension-shifting once again to establish the usual properties of the cup product (associativity, commutativity, unit); for these properties are known to hold in high dimensions. [Note: The unit for the cup product is the image in $\hat{H}^0(\Gamma, \mathbb{Z})$ of $1 \in H^0(\Gamma, \mathbb{Z}) = \mathbb{Z}$.]

Recall that the Tate group $\hat{H}^i$ for $i > 0$ could be interpreted in terms of homology. This resulted from the duality theory which allowed us to interpret the negative part of a complete resolution as the dual of an ordinary resolution. We can generalize this to Farrell cohomology, provided Γ is of type VFP . For simplicity we will assume that Γ is a virtual duality group, but the reader who has done exercises 2 and 3 of §2 should be able to treat the general case. [See also exercise 5 of §VIII.10.]

(3.5) Suppose that Γ is a virtual duality group with dualizing module D , and set $\hat{H}_*(\Gamma, M) = H_*(\Gamma, D \otimes M)$. Then there is an isomorphism $\hat{H}^i \approx \hat{H}_{n-1-i}$ for $i < -1$ and a monomorphism $\hat{H}^{-1} \rightarrow \hat{H}_n$. Moreover, the sequence

$$0 \longrightarrow \hat{H}^{-1}(\Gamma) \longrightarrow \hat{H}_n(\Gamma) \xrightarrow{\iota} \hat{H}_n(\Gamma')$$

is exact for any torsion-free subgroup $\Gamma' \subseteq \Gamma$ of finite index.

For let F be as in 2.5, and let $E = \Sigma^{n-1} \bar{F}$ (cf. remark at the end of §2). Then we have

$$\hat{H}^i(\Gamma, -) = H^i(\mathcal{H}om_{\Gamma}(F, -)) = H_{-i}(\bar{F} \otimes_{\Gamma} -) = H_{n-1-i}(E \otimes_{\Gamma} -).$$

Now the construction of E shows that there is a projective resolution Q of D which receives a chain map $E \rightarrow Q$ which is the identity in dimensions $\geq n$. Hence we have a map $H_j(E \otimes_{\Gamma} -) \rightarrow \text{Tor}_j^{\Gamma}(D, -)$ which is an isomorphism for $j > n$ and a monomorphism for $j = n$. Since $\text{Tor}_j^{\Gamma}(D, -) = \hat{H}_j(\Gamma, -)$ by III.2.2, this proves the first assertion of 3.5. The second assertion follows by a

dimension-shifting argument analogous to that used in 3.4; details are left to the reader.

We have not yet said anything about H^i for $0 \leq i \leq n-1$. This range of dimensions, which is vacuous if Γ is finite, is harder to understand than those treated in 3.4 and 3.5. We can say something about it, however, in view of the mapping cone construction in 2.5. For $\mathcal{H}om_{\Gamma}(F, M) = \bar{F} \otimes_{\Gamma} M$ is the mapping cone of a map $\Sigma^{-n}Q \otimes_{\Gamma} M \rightarrow \bar{P} \otimes_{\Gamma} M = \mathcal{H}om_{\Gamma}(P, M)$. We therefore have a long exact homology sequence, which takes the form

$$\cdots \rightarrow \tilde{H}_{n-i} \rightarrow H^i \rightarrow \hat{H}^i \rightarrow \tilde{H}_{n-1-i} \rightarrow H^{i+1} \rightarrow \cdots.$$

Since $H^{-1} = 0$ and $\tilde{H}_{-1} = 0$, we conclude:

(3.6) With the hypotheses and notation of 3.5, there is an exact sequence

$$0 \rightarrow \hat{H}^{-1} \rightarrow \tilde{H}_n \rightarrow H^0 \rightarrow \hat{H}^0 \rightarrow \tilde{H}_{n-1} \rightarrow H^1 \rightarrow \cdots \rightarrow \tilde{H}_0 \rightarrow H^n \rightarrow \hat{H}^n \rightarrow 0.$$

We can summarize 3.4–3.6 as follows. If Γ is a virtual duality group, then the Farrell cohomology functors $\hat{H}^i$ include: the cohomology functors H^i for $i > n$; the homology functors $\tilde{H}_i$ for $i > n$; a certain quotient of H^n ; a certain subfunctor of $\tilde{H}_n$; and n additional functors $\hat{H}^0, \dots, \hat{H}^{n-1}$, which are some sort of mixture of $\{H^i\}_{0 \leq i \leq n}$ and $\{\tilde{H}_i\}_{0 \leq i \leq n}$:

$$\begin{array}{ccccccccccccccc}
 & & & & & & H^0 & & \cdots & H^{n-1} & & H^n & H^{n+1} & H^{n+2} & \cdots \\
 & & & & & & \downarrow & & & \downarrow & & \downarrow & \parallel & \parallel & \\
 \cdots & \hat{H}^{-3} & \hat{H}^{-2} & \hat{H}^{-1} & & & \hat{H}^0 & \cdots & \hat{H}^{n-1} & \hat{H}^n & \hat{H}^{n+1} & \hat{H}^{n+2} & \cdots \\
 & \parallel & \parallel & \downarrow & \nearrow & \downarrow & & \downarrow & \nearrow & \downarrow & \parallel & \parallel & \\
 \cdots & \tilde{H}_{n+2} & \tilde{H}_{n+1} & \tilde{H}_n & & \tilde{H}_{n-1} & \cdots & \tilde{H}_0 & & & & & &
 \end{array}$$

EXERCISES

1. Let $\Gamma = \mathbb{Z} \times \mathbb{Z}_2$. Construct an explicit complete resolution and use it to compute $\hat{H}^*(\Gamma, \mathbb{Z})$. [Cf. exercise 1 of §2.]
2. Prove that the groups $\hat{H}^*(\Gamma)$ are torsion-groups. More precisely, they are annihilated by the integer d of IX.9.1. [Hint: Transfer.]

Remark. It is not known whether $\hat{H}^*(\Gamma)$ is annihilated by the integer m of IX.9.3. In view of the cup product structure on $\hat{H}^*(\Gamma)$, it would be enough to show that m annihilates the identity element $1 \in \hat{H}^0(\Gamma, \mathbb{Z})$.

3. (a) Define Farrell homology groups $\hat{H}_*(\Gamma)$.
 (b) If Γ is a virtual duality group with dualizing module D , show that $\hat{H}^i(\Gamma, M) \approx \hat{H}_{n-1-i}(\Gamma, D \otimes M)$ for all i . More precisely, show there is an element

$z \in \hat{H}_{n-1}(\Gamma, D)$ such that cap product with z is an isomorphism. [Hint: To construct an isomorphism $\psi: \hat{H}^i(\Gamma, M) \rightarrow \hat{H}_{n-1-i}(\Gamma, D \otimes M)$, use 3.5 and dimension-shifting; alternatively, take F and E as in the proof of 3.5 and show by homological algebra that there is a map $E \rightarrow F \otimes D$ which induces the desired isomorphism. Now let $z \in \hat{H}_{n-1}(\Gamma, D)$ be the image of $1 \in \hat{H}^0(\Gamma, \mathbb{Z})$ under ψ . Show that ψ is given by cap product with z (up to sign) as follows. The two composites

$$H^i(\Gamma, M) \rightarrow \hat{H}^i(\Gamma, M) \rightrightarrows \hat{H}_{n-1-i}(\Gamma, D \otimes M)$$

agree when $i = 0$ by naturality, hence they agree for all i by dimension-shifting. But then the two maps $\hat{H}^i(\Gamma, M) \rightrightarrows \hat{H}_{n-1-i}(\Gamma, D \otimes M)$ agree for large i and hence for all i .]

*(c) If D is $\mathbb{Z}$ -free in (b), show that cap product with the same $z \in \hat{H}_{n-1}(\Gamma, D)$ gives isomorphisms $\hat{H}_i(\Gamma, M) \approx \hat{H}^{n-1-i}(\Gamma, \text{Hom}(D, M))$.

*(d) State and prove analogues of (b) and (c) for arbitrary groups Γ of type VFP .

4. Let $\Gamma = \Gamma_1 \times \Gamma_2$ as in exercise 1 of §2. Derive, under suitable hypotheses, Künneth formulas relating $\hat{H}_(\Gamma)$ to $H_*(\Gamma_1)$ and $\hat{H}_*(\Gamma_2)$ and similarly for cohomology.

5. Let $1 \rightarrow \Gamma' \rightarrow \Gamma \rightarrow \Gamma'' \rightarrow 1$ be a short exact sequence of groups of finite virtual cohomological dimension.

(a) If Γ'' is torsion-free, show that there is a Hochschild-Serre spectral sequence (derived as in ordinary cohomology) of the form

$$E_2^{pq} = H^p(\Gamma'', \hat{H}^q(\Gamma')) \Rightarrow \hat{H}^{p+q}(\Gamma).$$

*(b) If Γ' is torsion-free, show that there is a spectral sequence

$$E_2^{pq} = \hat{H}^p(\Gamma'', H^q(\Gamma')) \Rightarrow \hat{H}^{p+q}(\Gamma).$$

[Method 1: Use the ordinary Hochschild-Serre spectral sequence and dimension-shifting. Method 2: Use a complete resolution of the form $F = F'' \otimes C$, where C is a finite dimensional complex of Γ -modules which is a Γ' -projective resolution of $\mathbb{Z}$.]

6. Define and study composition products $\hat{H}^ \otimes \hat{H}^* \rightarrow \hat{H}^*$ analogous to those of §VI.6. [The only tricky thing here is to prove the analogue of VI.6.1a. This is true, but the proof requires more work than in the case of finite groups.]

4 Equivariant Farrell Cohomology

Let Γ be a group of finite virtual cohomological dimension and let X be a *finite dimensional* Γ -CW-complex. As in Chapter VII, we then choose a complete resolution F for Γ and we set

$$\hat{H}_\Gamma^*(X; M) = H^*(\mathcal{H}om_\Gamma(F, C^*(X; M)))$$

for any Γ -module M . These *equivariant Farrell cohomology groups* have properties analogous to those of the ordinary equivariant cohomology groups (spectral sequences, etc.).

An important special case is that where X is taken to be proper and contractible (cf. VIII.11.1). We then obtain as in VII.7.10 a spectral sequence

$$(4.1) \quad E_1^{pq} = \prod_{\sigma \in \Sigma_p} \hat{H}^q(\Gamma_\sigma, M_\sigma) \Rightarrow \hat{H}^{p+q}(\Gamma, M),$$

where Σ_p is a set of representatives for the p -cells of $X \bmod \Gamma$. This spectral sequence relates the Farrell cohomology of Γ to the Tate cohomology of its finite subgroups. Note that the spectral sequence lives in the first and fourth quadrants, but there is no problem with convergence because $\dim X < \infty$. Indeed, the spectral sequence is concentrated in the vertical strip $0 \leq p \leq \dim X$, so $E_r = E_\infty$ as soon as $r > \dim X$.

The rest of this section will be devoted to establishing some properties of the spectral sequence 4.1 which will be needed in §6. For simplicity we will assume that X is an *ordered simplicial complex* with order preserving simplicial Γ -action. In addition we will assume that X^H is non-empty and connected for every finite subgroup $H \subseteq \Gamma$; we saw in §VIII.11 that such an X exists.

Note that the “order preserving” assumption implies that $M_\sigma = M$, i.e., that $\mathbb{Z}_\sigma = \mathbb{Z}$ with trivial Γ_σ -action, for each simplex σ . We may therefore suppress M from the notation in what follows.

Our first goal is to give a purely algebraic description of the left-hand edge $E_2^{0,q}$. For this we will need to compute the differential $d_1^{0,q}$. Recall that for any simplex σ of X and any $\gamma \in \Gamma$ we have a conjugation isomorphism

$$c(\gamma^{-1})^*: \hat{H}^*(\Gamma_\sigma) \xrightarrow{\cong} \hat{H}^*(\Gamma_{\gamma\sigma}),$$

denoted $u \mapsto \gamma u$ (cf. §III.8).

(4.2) Lemma. *Let X_p be the set of p -simplices of X . Then E_1^{pq} can be identified with the subgroup of $\prod_{\sigma \in X_p} \hat{H}^q(\Gamma_\sigma)$ consisting of those families $(u_\sigma)_{\sigma \in X_p}$ such that $\gamma u_\sigma = u_{\gamma\sigma}$ for all $\gamma \in \Gamma$, $\sigma \in X_p$. The differential d_1^{pq} is the restriction to this subgroup of the map*

$$d: \prod_{\sigma \in X_p} \hat{H}^q(\Gamma_\sigma) \rightarrow \prod_{\tau \in X_{p+1}} \hat{H}^q(\Gamma_\tau)$$

defined as follows: For any $\tau = (v_0, \dots, v_{p+1}) \in X_{p+1}$ with $v_0 < \dots < v_{p+1}$, let $\tau_i = (v_0, \dots, \hat{v}_i, \dots, v_{p+1})$, $i = 0, \dots, p+1$, and let $\rho_i: \hat{H}^q(\Gamma_{\tau_i}) \rightarrow \hat{H}^q(\Gamma_\tau)$ be the restriction map. Then d is given by

$$(u_\sigma) \mapsto \left(\tau \mapsto \sum_{i=0}^{p+1} (-1)^i \rho_i(u_{\tau_i}) \right).$$

(Note that $\Gamma_\tau \subseteq \Gamma_{\tau_i}$ because of the assumption that the Γ -action is order-preserving. Thus the definition of d makes sense.)

PROOF. Note first that

$$\prod_{\sigma \in X_p} \hat{H}^q(\Gamma_\sigma) = \prod_{\sigma \in \Sigma_p} \prod_{\gamma \in \Gamma/\Gamma_\sigma} \hat{H}^q(\Gamma_{\gamma\sigma}).$$

Since the conjugation action of Γ_σ on $\hat{H}^q(\Gamma_\sigma)$ is trivial (cf. III.8.3), it follows that there is a well-defined injection

$$\alpha: E_1^{pq} = \prod_{\sigma \in \Sigma_p} \hat{H}^q(\Gamma_\sigma) \rightarrow \prod_{\sigma \in X_p} \hat{H}^q(\Gamma_\sigma)$$

given by $(u_\sigma)_{\sigma \in \Sigma_p} \mapsto (\gamma u_\sigma)_{\sigma \in \Sigma_p, \gamma \in \Gamma/\Gamma_\sigma}$. The image of α is clearly equal to the set of families $(u_\sigma)_{\sigma \in X_p}$ satisfying $\gamma u_\sigma = u_{\gamma\sigma}$ for all $\sigma \in X_p$ and $\gamma \in \Gamma$, whence the first part of the proposition. The description of d_1 follows from the cohomology analogue of VII.8.1. For the convenience of the reader, however, we will give a direct proof.

Let F be a complete resolution for Γ . Then the injection

$$\alpha: E_1^{pq} \rightarrow \prod_{\sigma \in X_p} \hat{H}^q(\Gamma_\sigma)$$

is given in terms of cochains as follows: We have

$$\begin{aligned} E_1^{pq} &= \hat{H}^q(\Gamma, C^p(X; M)) \\ &= H^q(\mathcal{H}om_{\mathbb{Z}\Gamma}(F, \text{Hom}(X_p, M))) \\ &= H^q(\text{Hom}_\Gamma(X_p, \mathcal{H}om_{\mathbb{Z}}(F, M))), \end{aligned}$$

where the second equality comes from the ordering on X . An element $u \in E_1$ is therefore represented by a family $(c_\sigma)_{\sigma \in X_p}$ such that $c_\sigma \in \mathcal{H}om_{\mathbb{Z}}(F, M)$ and $\gamma c_\sigma = c_{\gamma\sigma}$ for all $\gamma \in \Gamma$ and $\sigma \in X_p$. Taking $\gamma \in \Gamma_\sigma$, it follows that $c_\sigma \in \mathcal{H}om_{\mathbb{Z}\Gamma_\sigma}(F, M)$. Hence c_σ represents an element $u_\sigma \in \hat{H}^q(\Gamma_\sigma, M)$, and one checks that $\alpha(u) = (u_\sigma)_{\sigma \in X_p}$. The differential d_1 is simply the map

$$\hat{H}^q(\Gamma, \delta): \hat{H}^q(\Gamma, C^p(X; M)) \rightarrow \hat{H}^q(\Gamma, C^{p+1}(X; M)),$$

where $\delta: C^p(X; M) \rightarrow C^{p+1}(X; M)$ is the coboundary operator. The lemma now follows at once from the definition of δ in terms of the face maps $X_{p+1} \rightarrow X_p$ and from the fact that the restriction map $\rho_i: \hat{H}^q(\Gamma_{\tau_i}) \rightarrow \hat{H}^q(\Gamma_\tau)$ is induced by the inclusion $\mathcal{H}om_{\mathbb{Z}\Gamma_{\tau_i}}(F, M) \hookrightarrow \mathcal{H}om_{\mathbb{Z}\Gamma_\tau}(F, M)$. $\square$

In particular, we can now easily calculate $E_2^{0,q} = \ker d_1^{0,q}$, and we obtain:

(4.3) Lemma. $E_2^{0,q}$ can be identified with the subgroup of $\prod_{v \in X_0} \hat{H}^q(\Gamma_v)$ consisting of those families $(u_v)_{v \in X_0}$ satisfying the following two conditions:

- (i) $\gamma u_v = u_{\gamma v}$ for any $\gamma \in \Gamma$, $v \in X_0$.
- (ii) if e is a 1-simplex of X with vertices v_0, v_1 , then u_{v_0} and u_{v_1} restrict to the same element of $\hat{H}^q(\Gamma_e)$.

Using now our hypothesis that X^H is non-empty and connected for H finite, we will deduce the desired algebraic description of $E_2^{0,q}$:

(4.4) Proposition. *Let $\mathfrak{F}$ be the set of finite subgroups of Γ . Then $E_2^{0,q}$ is isomorphic to the subgroup $\mathfrak{H}^q(\Gamma) \subseteq \prod_{H \in \mathfrak{F}} \hat{H}^q(H)$ consisting of those families $(u_H)_{H \in \mathfrak{F}}$ satisfying the following two conditions:*

- (i) $\gamma u_H = u_{\gamma H \gamma^{-1}}$ for all $\gamma \in \Gamma, H \in \mathfrak{F}$.
- (ii) If $H' \subseteq H$ then $\text{res}_H^H u_H = u_{H'}$.

(Note: Conditions (i) and (ii) could be combined into the single condition that the u_H are compatible with respect to “restriction maps” $\hat{H}^q(H_2) \rightarrow \hat{H}^q(H_1)$ induced by embeddings $H_1 \hookrightarrow H_2$ given by conjugation by elements of Γ .)

PROOF. Using the description of $E_2^{0,q}$ given in 4.3, we have an obvious map $\varphi: \mathfrak{H}^q(\Gamma) \rightarrow E_2^{0,q}$ given by $(u_H)_{H \in \mathfrak{F}} \mapsto (u_{\Gamma_v})_{v \in X_0}$. Since every $H \in \mathfrak{F}$ is contained in some Γ_v [because $X^H \neq \emptyset$], it follows from condition (ii) above that φ is injective. To prove that φ is surjective suppose $(u_v)_{v \in X_0}$ satisfies conditions (i) and (ii) of 4.3. Given $H \in \mathfrak{F}$, choose a vertex v such that $H \subseteq \Gamma_v$ and set $w_H = \text{res}_H^{\Gamma_v} u_v$. Since X^H is connected, condition (ii) of 4.3 shows that w_H is independent of the choice of v . It is easy to check that the resulting family $(w_H)_{H \in \mathfrak{F}}$ is in $\mathfrak{H}^q(\Gamma)$ and that its image under φ is (u_v) . $\square$

Next we wish to introduce a multiplicative structure into the spectral sequence 4.1. To simplify the notation, we will assume that the coefficient module M is a commutative ring R with trivial Γ -action. The simplicial cochain complex $C^*(X) = C^*(X; R)$ then has a cup product

$$C^*(X) \otimes C^*(X) \rightarrow C^*(X)$$

which is strictly associative and is commutative (in the graded sense) up to homotopy. Moreover, the product and the homotopy are defined canonically in terms of the structure of X as an ordered simplicial complex. [See the exercise below for a review of these facts.] They are therefore compatible with the Γ -action.

Now choose a complete resolution F with a diagonal map $\Delta: F \rightarrow F \hat{\otimes} F$. We then have a product

$$\begin{aligned} \mathcal{H}om_{\Gamma}(F, C^*(X)) \otimes \mathcal{H}om_{\Gamma}(F, C^*(X)) &\rightarrow \mathcal{H}om_{\Gamma}(F, C^*(X) \otimes C^*(X)) \\ &\rightarrow \mathcal{H}om_{\Gamma}(F, C^*(X)), \end{aligned}$$

where the first map is induced by Δ and the second map is induced by the cochain cup product $C^*(X) \otimes C^*(X) \rightarrow C^*(X)$. As we explained at the end of §VII.5, this product is compatible with the filtration defining the spectral sequence 4.1, hence there is an induced product

$$E_r \otimes E_r \rightarrow E_r$$

for $r \geq 1$. The following proposition summarizes the properties of this product which we will need:

(4.5) Proposition. (i) *The differential d_r is a derivation with respect to the product on E_r , i.e.,*

$$d_r(uv) = d_r(u) \cdot v + (-1)^{\deg u} u \cdot d_r(v).$$

(ii) *The product on $E_{r+1} = H(E_r)$ is obtained from the product on E_r by passage to homology.*

(iii) *The product on E_1 is the composite*

$$\begin{aligned} \hat{H}^q(\Gamma, C^p(X)) \otimes \hat{H}^{q'}(\Gamma, C^{p'}(X)) &\rightarrow \hat{H}^{q+q'}(\Gamma, C^p(X) \otimes C^{p'}(X)) \\ &\rightarrow \hat{H}^{q+q'}(\Gamma, C^{p+p'}(X)) \end{aligned}$$

where the first map is the usual cup product in $\hat{H}^*(\Gamma, -)$ and the second map is induced by the cup product in $C^*(X)$.

(iv) *The product on E_r is associative for $r \geq 1$ and commutative for $r \geq 2$.*

(v) *The product on E_∞ is compatible with the usual product on $\hat{H}^*(\Gamma, R)$ under the identification of E_∞ with $\text{Gr } \hat{H}^*(\Gamma, R)$.*

(vi) *The isomorphism $E_2^{0,*} \approx \mathfrak{H}^*(\Gamma)$ of 4.4 is a ring isomorphism.*

PROOF. (i) and (ii) follow from the fact that the product on E_r is induced by the product on $\mathcal{H}om_\Gamma(F, C^*(X))$. (iii) is immediate from the definitions. In view of (iii) and properties of the ordinary cup product, the product on E_1 is associative and is commutative up to homotopy (with respect to the differential d_1). (iv) now follows from (ii). The product on E_∞ is obviously compatible with the product on the abutment $H^*(\mathcal{H}om_\Gamma(F, C^*(X)))$. But the identification of the abutment with $\hat{H}^*(\Gamma)$ is induced by a chain map $\mathcal{H}om_\Gamma(F, R) \rightarrow \mathcal{H}om_\Gamma(F, C^*(X))$ which is easily seen to be a ring homomorphism, whence (v). Finally, to prove (vi) it suffices to show that the map $\alpha: E_1^{0,*} \rightarrow \prod_{v \in X_0} \hat{H}^*(\Gamma_v)$ defined in the proof of 4.2 is a ring homomorphism. Consider the v -component of α ,

$$E_1^{0,*} = \hat{H}^*(\Gamma, C^0(X)) \rightarrow \hat{H}^*(\Gamma_v).$$

It is given on the cochain level by a map

$$\mathcal{H}om_\Gamma(F, C^0(X)) \rightarrow \mathcal{H}om_{\Gamma_v}(F, R),$$

which in turn is induced by the ring homomorphism $C^0(X) = \text{Hom}(X_0, R) \rightarrow R$ given by evaluation at v . This cochain map is easily seen to be a ring homomorphism, so (vi) is proved. $\square$

As a simple application of this multiplicative structure we will prove a result due to Quillen [1971]. Fix a prime p and consider the ring $\mathfrak{H}^*(\Gamma, \mathbb{Z}_p)$ defined in 4.4. There is an obvious homomorphism

$$\rho: \hat{H}^*(\Gamma, \mathbb{Z}_p) \rightarrow \mathfrak{H}^*(\Gamma, \mathbb{Z}_p)$$

given by the restriction maps $\hat{H}^*(\Gamma) \rightarrow \hat{H}^*(H)$ ($H \in \mathfrak{F}$).

(4.6) Proposition. *The map ρ has the following two properties:*

- (i) *Every element of $\ker \rho$ is nilpotent.*
- (ii) *For any $u \in \mathfrak{H}^*(\Gamma, \mathbb{Z}_p)$ there is an integer $k \geq 0$ such that $u^{p^k} \in \text{im } \rho$.*

(Thus ρ is an isomorphism “up to p -th powers.” Following Quillen, we say that ρ is an F -isomorphism.)

PROOF. Note first that ρ is simply the edge homomorphism

$$\hat{H}^*(\Gamma) \rightarrow E_\infty^{0,*} \hookrightarrow E_2^{0,*} \approx \mathfrak{H}^*(\Gamma)$$

associated to the spectral sequence 4.1. This is easily seen by direct definition-checking (see also exercise 1 of §VII.7). Recall that the construction of the spectral sequence gives us a filtration

$$\hat{H}^*(\Gamma) = F^0 \hat{H}^*(\Gamma) \supseteq F^1 \hat{H}^*(\Gamma) \supseteq \dots$$

which is compatible with the cup product (i.e., $F^s F^t \subseteq F^{s+t}$). Since the kernel of the edge homomorphism above is precisely $F^1 \hat{H}^*(\Gamma)$, it follows that any $z \in \ker \rho$ satisfies $z^k \in F^k \hat{H}^*(\Gamma)$. But $F^k \hat{H}^*(\Gamma) = 0$ for $k > \dim X$, so z is nilpotent. Next let $u \in E_2^{0,*}$. I claim that $d_2(u^p) = 0$. This follows from anti-commutativity if p and $\deg u$ are both odd (in which case $u^2 = 0$); otherwise, it follows from (i) and (iv) of 4.5, which give

$$d_2(u^p) = pu^{p-1}d_2(u) = 0$$

since the coefficient module is $\mathbb{Z}_p$. Thus $u^p \in E_3^{0,*}$. Iterating this argument, we find that $u^{p^k} \in E_{k+2}^{0,*}$, hence $u^{p^k} \in E_\infty^{0,*} = \text{im } \rho$ as soon as $k + 2 > \dim X$. $\square$

Remark. Quillen actually proved a much stronger result. Namely, he proved the analogue of 4.6 with $\mathfrak{H}^*(\Gamma)$ defined in terms of the finite subgroups of Γ which are elementary abelian p -groups.

EXERCISE

Let K be an ordered simplicial complex and let $C(K)$ be its chain complex, with one basis element for every $(n+1)$ -tuple $(v_0, \dots, v_n)$ of vertices such that $\{v_0, \dots, v_n\}$ is an n -simplex with $v_0 < \dots < v_n$. The Alexander-Whitney diagonal map $\Delta: C(K) \rightarrow C(K) \otimes C(K)$ is defined by

$$\Delta(v_0, \dots, v_n) = \sum_{p=0}^n (v_0, \dots, v_p) \otimes (v_p, \dots, v_n).$$

Verify that Δ is a chain map and that the induced cochain cup product is associative and is commutative up to (canonical) homotopy. [Hint: The homotopy commutativity can be proved by an acyclic models argument, which is most easily carried out as follows: Because of the ordering on the vertices of K , a simplex of K can be regarded as a singular simplex in the geometric realization of K . Thus $C(K)$ is a subcomplex of the singular

chain complex $C^{\text{sing}}(K)$. Now look at the standard proof via acyclic models that the Alexander–Whitney map on $C^{\text{sing}}(K)$ is naturally homotopy commutative, and observe that the homotopy can be taken to map $C(K)$ into $C(K) \otimes C(K)$.]

5 Cohomologically Trivial Modules

We continue to assume that $\text{vcd } \Gamma = n < \infty$.

(5.1) Lemma. *If M is a Γ -module such that $\hat{H}^*(G, M) = 0$ for every finite subgroup $G \subseteq \Gamma$, then $\hat{H}^*(\Gamma, M) = 0$.*

PROOF. Consider the spectral sequence 4.1, and recall that X can be chosen so that $M_\sigma = M$ for all σ . Our hypothesis implies that $E_1 = 0$, hence the abutment $\hat{H}^*(\Gamma, M)$ is also zero. $\square$

We will say that M is *cohomologically trivial* if $\hat{H}^*(\Gamma', M) = 0$ for every subgroup $\Gamma' \subseteq \Gamma$. Applying 5.1 to each Γ' , we obtain:

(5.2) Proposition. *A Γ -module is cohomologically trivial if and only if it is cohomologically trivial as a G -module for every finite subgroup $G \subseteq \Gamma$.*

Thus the theory of cohomologically trivial modules is reduced to the case where Γ is finite. In particular, we can use this to prove the following generalization of Rim's theorems VI.8.10 and VI.8.12:

(5.3) Theorem. *The following conditions on a Γ -module M are equivalent:*

- (i) M is cohomologically trivial.
- (ii) $\text{proj dim}_{\mathbb{Z}\Gamma} M \leq n + 1$.
- (iii) $\text{proj dim}_{\mathbb{Z}\Gamma} M < \infty$.

If these conditions hold and M is $\mathbb{Z}$ -free, then $\text{proj dim}_{\mathbb{Z}\Gamma} M \leq n$.

PROOF. We have (ii) $\Rightarrow$ (iii) trivially, and (iii) $\Rightarrow$ (i) exactly as in the proof of VI.8.12. To prove (i) $\Rightarrow$ (ii), assume first that M is $\mathbb{Z}$ -free, in which case we will prove that $\text{proj dim}_{\mathbb{Z}\Gamma} M \leq n$. According to VIII.2.1, it will suffice to show that $\text{Ext}_{\Gamma}^{n+1}(M, N) = 0$ for every Γ -module N . Moreover, a glance at the proof of VIII.2.1 shows that we only need to consider modules N which are $\mathbb{Z}$ -free. In this case we know from VI.8.11 that the Γ -module $\text{Hom}(M, N)$ is cohomologically trivial over every finite subgroup of Γ , hence $\text{Hom}(M, N)$ is cohomologically trivial by 5.2. We therefore have

$$\begin{aligned} \text{Ext}_{\Gamma}^{n+1}(M, N) &= H^{n+1}(\Gamma, \text{Hom}(M, N)) && \text{by III.2.2} \\ &= \hat{H}^{n+1}(\Gamma, \text{Hom}(M, N)) && \text{by 3.4} \\ &= 0. \end{aligned}$$

If M is now cohomologically trivial but not $\mathbb{Z}$ -free, choose a short exact sequence $0 \rightarrow M' \rightarrow P \rightarrow M \rightarrow 0$ with P projective. Then M' is cohomologically trivial and $\mathbb{Z}$ -free, so $\text{proj dim } M' \leq n$ by what we just proved. Therefore $\text{proj dim } M \leq n + 1$. $\square$

EXERCISE

Show that the following conditions are equivalent:

- (i) M is cohomologically trivial.
- (ii) For every subgroup $\Gamma' \subseteq \Gamma$ there is an integer m such that $H^i(\Gamma', M) = 0$ for $i > m$.
- (iii) $\hat{H}_*(\Gamma', M) = 0$ for every subgroup $\Gamma' \subseteq \Gamma$.
- (iv) For every subgroup $\Gamma' \subseteq \Gamma$ there is an integer m such that $H_i(\Gamma', M) = 0$ for $i > m$.

6 Groups with Periodic Cohomology

A group Γ of finite virtual cohomological dimension is said to have *periodic cohomology* if for some $d \neq 0$ there is an element $u \in \hat{H}^d(\Gamma, \mathbb{Z})$ which is invertible in the ring $\hat{H}^*(\Gamma, \mathbb{Z})$. Cup product with u then gives a *periodicity isomorphism*

$$\hat{H}^i(\Gamma, M) \approx \hat{H}^{i+d}(\Gamma, M)$$

for any Γ -module M and any $i \in \mathbb{Z}$. Similarly, we say that Γ has *p-periodic cohomology* (where p is a prime) if the p -primary component $\hat{H}^*(\Gamma, \mathbb{Z})_{(p)}$, which is itself a ring (cf. exercise 2a of §VI.5), contains an invertible element of non-zero degree d . We then have

$$\hat{H}^i(\Gamma, M)_{(p)} \approx \hat{H}^{i+d}(\Gamma, M)_{(p)}.$$

As in exercise 2b of §VI.5 (see also exercise 2 of §3), we have

$$\hat{H}^*(\Gamma, \mathbb{Z}) \approx \prod_p \hat{H}^*(\Gamma, \mathbb{Z})_{(p)},$$

where p ranges over the primes such that Γ has p -torsion. Since this is a finite direct product of rings, it is clear that Γ has periodic cohomology if and only if Γ has p -periodic cohomology for every prime p . We will therefore restrict our attention to p -periodicity. Our goal, as in the previous section, will be to reduce to the case where Γ is finite, which we already understand by §VI.9. First of all, it is sometimes convenient to use $\hat{H}^*(\Gamma, \mathbb{Z}_p)$ instead of $\hat{H}^*(\Gamma, \mathbb{Z})_{(p)}$. This is justified by:

(6.1) Proposition. Γ has p -periodic cohomology if and only if $\hat{H}^*(\Gamma, \mathbb{Z}_p)$ contains an invertible element of non-zero degree.

The proof will be based on a “Bockstein” argument. Let $H(k) = \hat{H}^*(\Gamma, \mathbb{Z}_{p^k})$. For any k, l there is a canonical short exact sequence $0 \rightarrow \mathbb{Z}_{p^l} \rightarrow \mathbb{Z}_{p^{k+l}} \rightarrow \mathbb{Z}_{p^k} \rightarrow 0$, which yields a cohomology exact sequence

$$(6.2) \quad \cdots \rightarrow H(l) \rightarrow H(k+l) \xrightarrow{\alpha_{k+l,k}} H(k) \xrightarrow{\beta_{k,l}} H(l) \rightarrow \cdots$$

(6.3) **Lemma.** *If $k \geq l$ then $\beta_{k,l}$ is a derivation, i.e.,*

$$\beta_{k,l}(uv) = \beta_{k,l}(u) \cdot v + (-1)^{\deg u} \cdot \beta_{k,l}(v).$$

(The right-hand side makes sense and is in $H(l)$ because we have a cup product $H(k) \otimes H(l) \rightarrow H(l)$ for $k \geq l$.)

PROOF. Choose a complete resolution F and a diagonal map $F \rightarrow F \hat{\otimes} F$. It is easy to see that $\beta_{k,l}$ can be computed as follows: Given $u \in H(k)$, choose a cochain $c \in \mathcal{H}om_{\Gamma}(F, \mathbb{Z})$ whose reduction mod p^k is a cocycle representing u ; then δc is divisible by p^k , and $\delta c/p^k$ is a cocycle whose reduction mod p^l represents $\beta_{k,l}(u)$. The derivation property of $\beta_{k,l}$ now follows from the corresponding property of δ . $\square$

(6.4) **Lemma.** *For any $u \in H(1)$ and any $k \geq 1$, $u^{p^k} \in \text{im } \alpha_{k+1,1} = \ker \beta_{1,k}$.*

PROOF. I claim first that for any $u \in H(k)$, we have $u^p \in \ker \beta_{k,1} = \text{im } \alpha_{k+1,k}$. This follows from the fact that $\beta_{k,1}$ is a derivation whose target $H(1)$ is annihilated by p (cf. proof of 4.6). Thus $\alpha_{k+1,1}$ is the composite of ring homomorphisms

$$H(k+1) \xrightarrow{\alpha_{k+1,k}} H(k) \rightarrow \cdots \rightarrow H(1),$$

each of which has the property that p -th powers are in the image. The lemma follows at once. $\square$

Let $H(\infty) = \hat{H}^*(\Gamma, \mathbb{Z}_{(p)})$. Note that we still have an exact sequence of the form 6.2 if $l = \infty$ (and $k < \infty$); this is obtained by taking p -primary components in the cohomology exact sequence associated to

$$0 \rightarrow \mathbb{Z} \xrightarrow{p^k} \mathbb{Z} \rightarrow \mathbb{Z}_{p^k} \rightarrow 0.$$

(6.5) **Lemma.** *For sufficiently large k , $\text{im } \alpha_{k,1} = \text{im } \alpha_{\infty,1}$.*

PROOF. We know (exercise 2 of §3) that $H(\infty)$ is annihilated by p^k for some k . The sequence 6.2 with $l = \infty$ therefore yields an injection $\alpha_{\infty,k}: H(\infty) \hookrightarrow H(k)$ for large k . Now it is easy to see that $\beta_{1,k} = \alpha_{\infty,k}\beta_{1,\infty}$, so $\ker \beta_{1,k} = \ker \beta_{1,\infty}$ for large k . But $\ker \beta_{1,k} = \text{im } \alpha_{k+1,1}$ and $\ker \beta_{1,\infty} = \text{im } \alpha_{\infty,1}$. $\square$

(6.6) **Lemma.** *The map $\alpha = \alpha_{\infty,1}: H(\infty) \rightarrow H(1)$ has the following two properties:*

- (i) *Every element of $\ker \alpha$ is nilpotent.*
- (ii) *For any $u \in H(1)$ there is an integer k such that $u^{p^k} \in \text{im } \alpha$.*